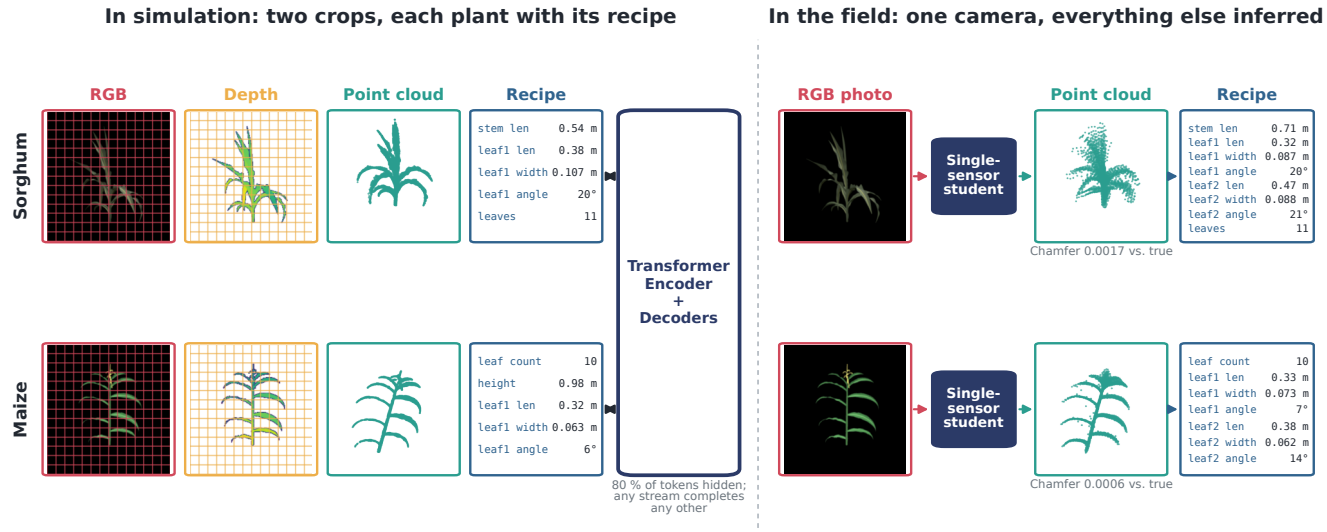
Procedural Reconstruction of Slender Plants from Any Field Modality

Anonymous CVPR submission

Paper ID *****



_1_plant_0004_04 (raw files in paper/data); right: real outputs, sorghum zero-shot (pretrained arm), maize distilled; no ground truth shown. RGB shown as the current renders (brightened sorghum, background-free maize); each m

Figure 1. **Any sensor in, procedural plant out.** From any subset of field-practical modalities (a single photograph, a depth frame, or a point cloud truncated by row neighbors), CROPCODE recovers the full procedural representation of the plant: a kilobyte-scale, physiologically interpretable parameter set that can be re-rendered to complete geometry, used as simulation input, and used for trait analytics at field scale.

Abstract

Editable, biologically interpretable 3D models of field-grown crops support high-throughput phenotyping, in silico *breeding trials, and agronomic digital twins, but recovering such models from field sensor data remains difficult: plant organs are slender, neighboring plants occlude one another, and a field campaign rarely yields more than a single photograph, a noisy depth frame, or a partial point cloud per plant. Slender geometry under occlusion is a regime in which learned 3D completion and Chamfer-based evaluation both fail. In this work, we present* CROPCODE*, a multimodal masked-autoencoder framework that recovers the complete procedural representation of a plant, the generator parameters encoding leaf count, insertion heights, blade angles, and surface geometry, from any subset of RGB, depth, and point-cloud input in a single forward pass. The model is pretrained on 15,000 procedurally generated plants per species (maize and sorghum), fine-tuned on rows in which simulated neighbors occlude the target, and distilled from a full-modality teacher into single-sensor students. Distillation lowers the cross-modal generation loss by 30% in sorghum and 44% in maize, and halves the sorghum parameter error from a point cloud alone. Training on rows removes 92% (maize) and 85% (sorghum) of the error that neighbors add for a model trained on isolated plants. Controlled ablations in both species show that exposing most of the procedural parameters to the encoder degrades phenotype recovery, while using them only as a reconstruction target degrades the sorghum representation and improves the maize one. We further show that Chamfer distance and linear probes on the frozen representation rank input modalities differently on slender geometry, and we adopt the probe as the deciding metric. On real field scans with independently reconstructed ground truth,* CROPCODE *matches the accuracy of a staged annotation pipeline and reduces the time required for procedural reconstruction from hours of manual specification, or*

minutes of pipeline computation, to milliseconds per plant. We will release FIELDPROC*, the 30,000-plant, two-species benchmark behind these experiments, which pairs multimodal field captures under simulated row occlusion with exact procedural ground truth.*

1. Introduction

Accurate three-dimensional models of individual crop plants enable high-throughput phenotyping of morphological traits such as leaf angle, internode length, and canopy architecture, and they are the basis of emerging agricultural digital twins in which breeding trials run *in silico* [19, 25]. The representation best suited to these uses is procedural: a compact set of biologically interpretable parameters that a generator expands into complete, editable, simulation-ready geometry [20, 35]. A procedural plant occupies a few kilobytes where a laser scan occupies megabytes, every parameter corresponds to a trait used by agronomists, and the model can be edited, re-rendered, and perturbed in ways that unstructured geometry does not support. Field crops also instantiate a challenge of general interest in 3D computer vision: recovering a parametric model of a thin, heavily occluded object from partial multimodal observations.

Yet recovering procedural structure from field-acquired sensor data remains a significant challenge, and every existing route presupposes a rich, clean observation. Manual parameter specification takes hours per plant [20]. Staged reconstruction pipelines automate the process; a recent system annotates laser scans with a vision-language model and fits a procedural descriptor to 5.4 mm median accuracy, but it requires a complete scan of an isolated plant and roughly ten minutes of computation per plant [18]. Per-scene optimization methods search generator parameters against multi-view imagery [52], and recent learned models infer parametric plants, but only from a clean, complete point cloud [8, 14]. A phenotyping campaign produces the opposite: thousands of plants per day, each observed by whichever sensor was practical (a camera, a depth frame, or a partial scan).

In addition, crop plants are slender: a maize blade is centimeters wide, millimeters thick, and up to a meter long. In a planted row, each plant is further occluded by its neighbors, whose leaves interleave with the target; in our field simulations, a third of the target's pixels and two thirds of the scene's point cloud belong to neighboring plants. This regime violates the assumptions of learned 3D completion: the standard benchmarks contain compact, largely convex objects and no slender-under-occlusion category [49, 51], and Chamfer distance, the default metric, is structurally insensitive to thin sheets, so it can reward reconstructions that miss entire blades [42, 47]. Occlusion-aware completion of thin plant organs has begun to address the input side of this regime, but its output is a completed point cloud rather than structure [27]. To our knowledge, no prior method recovers a procedural, re-renderable representation of slender crop organs from a single field-practical observation (a photograph, a depth frame, or a partial point cloud) under row-neighbor occlusion: methods that infer plant parameters require clean, complete scans of isolated plants [8, 14], and methods that tolerate field occlusion complete unstructured geometry instead [27].

In this work, we present CROPCODE, a framework that makes this inverse problem tractable by using the procedural generator itself as the prior. We pretrain a four-stream masked autoencoder [3, 10, 21, 31] over RGB, depth, point clouds, and tokenized procedural parameters of 15,000 generated plants per species, maize and sorghum, and fine-tune it on simulated rows in which neighboring leaves bend on contact. Because random modality subsets are masked during pretraining, a single model learns to complete any subset at inference. A full-modality teacher is then distilled [22] into single-sensor students for deployment. Distillation lowers the cross-modal generation loss by 30% in sorghum and 44% in maize, and it halves the sorghum parameter error from a point cloud alone. Row fine-tuning against the target plant behind its neighbors removes most of the error that the neighbors add: 92% in maize and 85% in sorghum.

We also address a design question that arises for any model with access to generative structure: where should the parameter stream enter? In our controlled ablations, a parameter stream that is largely visible to the encoder harms the representation in both species. With the shared token budget, adding the parameter stream to the encoder degrades the frozen representation on every sorghum phenotype target, and on maize the paired comparison is a draw. Control arms that restore the sensor-token budget separate the species: in sorghum the loss persists, even when the parameters serve only as a reconstruction target for the decoder, while in maize the arms that see 20 % of the parameters or only reconstruct them improve every trait. Supervising parameters as a generation target, however, lets the distilled generalist outperform a parameter-only specialist at parameter regression (first-release result: no final-data specialist was trained). One interpretation, consistent with modality competition [34] and negative transfer [48], is that a stream that already contains the answer suppresses learning in the streams that must provide the trait information at test time.

Adding depth to the photograph and the point cloud lowers Chamfer distance by 28% in sorghum and 25% in maize, whereas a linear probe on the frozen representation, read out on phenotypic traits, finds little or no gain from depth and identifies RGB as the main source of trait information. On slender geometry the two metrics measure different quan-

tities, surface proximity and recovered structure, and we adopt the probe as the deciding metric for modality decisions [6, 47].

We claim that fragmentary field sensing suffices to recover a complete, interpretable procedural plant, provided the generator is used for both the training data and the output representation. We validate this claim on real field scans of maize and sorghum with independently reconstructed ground truth [18, 25], including rows in which neighbors occlude the target, demonstrating accuracy matching staged annotation pipelines while running in milliseconds per plant. The key contributions of this work are:

1. **Task and framework.** The first formulation and system, to our knowledge, for recovering procedural plant representations of slender, row-occluded field crops from *any subset* of RGB, depth, and point-cloud input in one amortized forward pass, with full-modal to single-sensor distillation (generation loss -30% in sorghum and -44% in maize; sorghum parameter error from a point cloud alone halved), and row fine-tuning that removes most of the error added by neighboring plants (92% in maize, 85% in sorghum).
2. **A design rule for structured modalities.** Controlled, budget-matched, two-species evidence that a parameter stream largely visible to the encoder degrades the sensor representation in both species, while the value of the parameters as a reconstruction target depends on the generator (it degrades the sorghum representation and improves the maize one); we expect this guidance to transfer to any multimodal model with access to privileged generative structure.
3. **An evaluation protocol for slender geometry.** A demonstration that Chamfer distance and downstream phenotype probes disagree systematically on slender organs, and a frozen-latent probe protocol that resolves modality decisions where Chamfer is insensitive.
4. **A benchmark.** FIELDPROC, a two-species procedural field benchmark of 30,000 plants spanning diverse phenotypes, generator configurations, and row occlusion severities, with paired multimodal captures, physically simulated row occlusion, and exact parameter ground truth, together with a real-field evaluation split carrying reconstructed procedural ground truth across the genotypes of a diversity panel [25]. No existing dataset combines these elements (Tab. 2) [41, 51]. We will release the data, generation code, and evaluation scripts.

2. Related Work

Recovering procedural plants. Inverse procedural modeling has a long history for woody plants: parameters of tree generators have been recovered by stochastic search [43], by Markov chain Monte Carlo inference [44], and by inferring L-systems from point clouds [17, 28]. These methods optimize or infer per instance. Recent systems address crops specifically. Demeter learns a parametric morphology model from soybean point clouds [8], Ghrer et al. infer grammar-style plant representations from 3D scans [14], CropCraft fits a procedural crop model by Bayesian optimization against multi-view imagery [52], and a recent staged pipeline reconstructs procedural maize from terrestrial laser scans through vision-language-model annotation and differentiable NURBS refinement [9, 18]. Each of these methods presupposes a single rich observation, a clean and complete point cloud or a multi-view capture of an isolated plant, and none addresses missing modalities or inter-plant occlusion.

Programs from observations. Outside agriculture, decoding parametric programs from observations is an established pattern: neural shape programs [45], diffusion-based inverse procedural content generation from a single image [53], text-conditioned CAD generation [24], and cadrille, which maps images, point clouds, or text to executable CAD models [26]. These systems operate on rigid objects observed cleanly and in isolation. Our setting differs in the geometry (deformable, slender organs), in the input (noisy, occluded field sensors), and in a design question that none of them raises: whether the program parameters should be visible to the encoder at all.

Completion and slender geometry. Learned point cloud completion has progressed from folding-based decoders [51] to geometry-aware transformers [5, 49, 50], trained on benchmarks that contain compact, largely convex objects and no slender-under-occlusion category. Completion of thin plant organs under occlusion has emerged recently [27], and reconstruction of thin structures from clean multi-view data is an active topic [13, 38], but in both lines the output is unstructured geometry rather than an editable model. Chamfer distance, the standard completion metric, is insensitive to thin sheets and can be reduced by collapsing predictions onto bulk structure [30, 42, 47]; we therefore evaluate with trait probes alongside geometric metrics (Sec. 4.7).

Multimodal masked pretraining. Masked autoencoding [21] extends to multiple modalities in MultiMAE [3], to any-to-any generation over tokenized modalities in 4M [4, 31], to point clouds in Point-MAE [33], and to combined RGB, depth, and point cloud inputs in EmbodiedMAE, on which our architecture is based [10]. Robustness to missing modalities is a recognized concern [40], and distillation from multi-sensor teachers to single-sensor students is established for discriminative tasks [22]. We extend these methods in two ways: we add a structured generative modality, encoded with continuous number tokens [15], and

we distill a generative full-modality teacher for structured parameter regression.

Learning from procedural data. Pretraining on purely synthetic data is viable, from abstract fractals [23] to procedurally generated scenes [16, 39], with domain randomization reducing the gap to real data [46] and with the value of synthetic data known to depend on the downstream task [11]. Our setting is the favorable case for this strategy: the procedural generator defines the output representation, spans the morphology of the target crop, and is paired with real field evaluation data [25]. Our results add one caveat to this strategy: the generator's parameters are valuable as data, but exposing most of them to the encoder reduces representation quality in both species, and their value as a reconstruction target depends on the generator (Sec. 4.6).

3. Method

In this section, we introduce CROPCODE, a unified multimodal masked-autoencoder framework designed to recover editable procedural plant models from fragmentary and occluded field observations. We begin by formulating the procedural inverse reconstruction problem (Sec. 3.1). We then describe our synthetic field-data engine, detailing how procedurally generated crop plants and physically simulated row occlusion scenes are created (Sec. 3.2). Next, we explain the four-stream tokenization, dynamic Dirichlet masking schedule, and multi-term reconstruction loss functions (Sec. 3.3). Finally, we present our teacher–student distillation strategy for deployment on single sensors (Sec. 3.4) and detail our experimental evaluation and domain transfer protocols (Sec. 3.5). An overall schematic of the framework is provided in Fig. 2.

3.1. Problem formulation: inverse procedural reconstruction

A procedural plant generator G is a generative function that maps a continuous vector of parameter attributes $\theta \in \mathbb{R}^{d_\theta}$ to an editable, full-resolution 3D triangular mesh $G(\theta)$. The vector θ acts as a complete descriptor of the generated plant: it encodes macro-level plant structure (such as stem height and internode spacing) as well as organ-level morphology for every leaf slot (including insertion height, blade length and width, and insertion, roll, and azimuth angles).

In real-world agricultural field settings, sensors yield partial, noisy, and occluded observations. Let $\mathcal{M} = \{\mathrm{rgb}, \mathrm{d}, \mathrm{pc}\}$ represent the universe of available field sensor modalities: a 2D RGB photograph, a single depth frame, and an unstructured 3D point cloud, respectively. In any given field campaign, a user may acquire only an incomplete subset $S \subseteq \mathcal{M}$ (for example, a single photograph taken from a mobile device or a partial point cloud captured by a tractor-mounted depth camera). Furthermore, neighboring plants in a crop row routinely interleave their leaves, causing heavy inter-plant occlusion.

Parameter / Configuration	Maize	Sorghum
Total generated plants	15,000	15,000
Split (train / val / test)	10,500 / 2,250 / 2,250	
Views per plant	10 (random sphere)	10 (fixed rig)
Camera distance	box-fitted	2.5 m
Per-view point cloud	8,192 pts (visible)	complete plant cloud
Parameter dim. d_θ	$14{+}14{\times}28{=}406$	$9{+}5{\times}24{=}129$
Leaf count range	3–22	4–35
Plant height range	0.62–1.44 m	0.19–1.74 m (stem)
Row scenes × orientations	$15{,}000 \times 10$	$15{,}000 \times 10$
Row camera	level rover, 1.0 m	level rover, 2.5 m
Row cloud (input / target)	24,576 / 8,192 pts	
Real-field split	MaizeField3D	Field scans

Table 1. **FIELDPROC dataset configuration.** Parameter dimensions include macro plant-level traits plus K structured leaf parameter slots. Every row scene pairs a target main plant with two adjacent row neighbors spaced at 0.15 m stem intervals.

Our primary goal is to learn an amortized feed-forward inverse mapping f_ϕ:

$$f_\phi(\{x^m : m \in S\}) = \left(\hat{\theta}, \{\hat{x}^m : m \in \mathcal{M}\}\right), \quad (1)$$

where ϕ denotes trainable network parameters, $\hat{\theta}$ is the predicted procedural description, and $\hat{x}^m$ represents the completed 3D geometry and visual modalities.

Intuition behind the formulation. Traditional 3D completion methods attempt to complete missing geometry by outputting raw point clouds or voxel grids. However, slender crop organs under row occlusion present an ill-posed inverse problem: when a leaf blade is hidden behind a neighbor, raw geometric completion can only guess point locations. In contrast, by predicting the underlying procedural parameters $\hat{\theta}$, CROPCODE uses the structure of the generator G as a strong generative prior. Predicting $\hat{\theta}$ guarantees that the output mesh $G(\hat{\theta})$ is structurally complete, editable, and interpretable in terms of physiological traits used by agronomists. Furthermore, while per-plant numerical optimization pipelines require minutes to hours per scan, our feed-forward network f_ϕ executes the inversion in a single forward pass taking milliseconds.

3.2. A procedural field-data engine: FIELDPROC

To train and evaluate CROPCODE, we construct FIELDPROC, a synthetic benchmark pairing exact procedural ground-truth parameters with multimodal field observations under realistic row neighbor occlusion (Tab. 1).

Procedural plant generation. We build generators for two major agricultural crop species with distinct structural

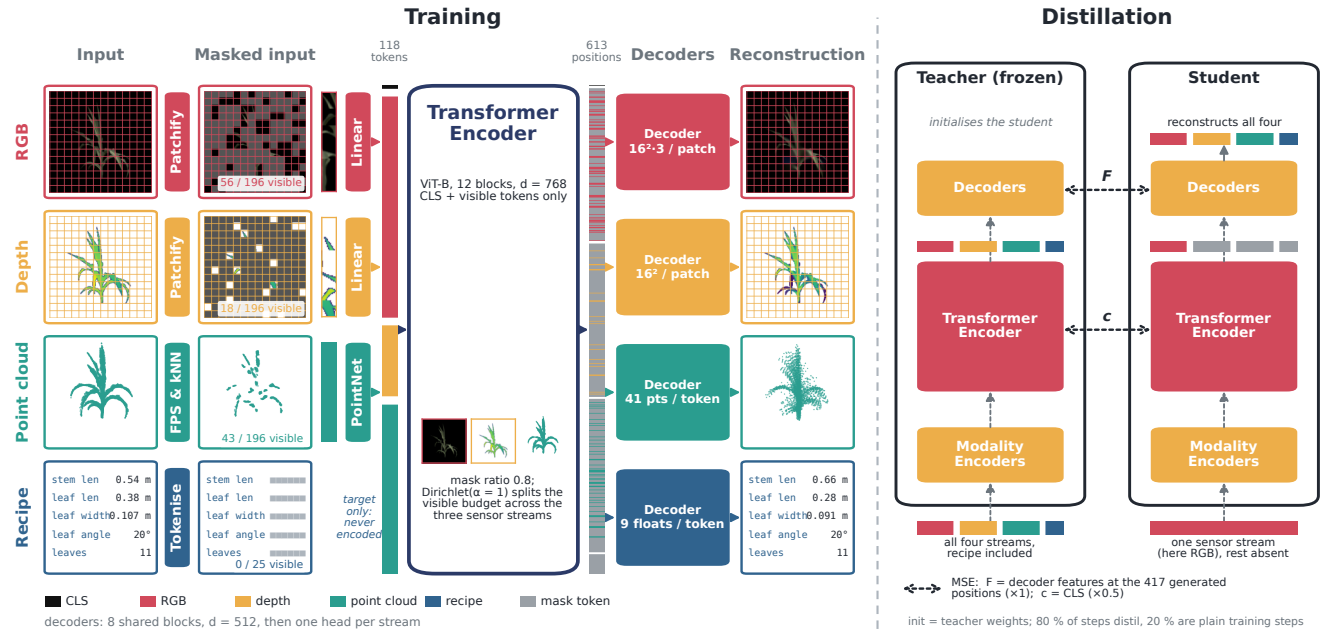


Figure 2. **CROPCODE framework overview.** (a) **Four-stream pretraining**: Tokenized streams for RGB images, metric depth frames, 3D point clouds, and continuous procedural parameters enter a shared ViT encoder under a dynamic Dirichlet masking budget. Modality-specific decoder heads reconstruct all target streams. During pretraining, procedural parameters act as both masked encoder input and decoder target. (b) **Teacher–student distillation**: A frozen full-modality teacher model supervises single-sensor and dual-sensor student models, transferring privileged multi-sensor latents to single-forward-pass field models.

dynamics: maize (*Zea mays*) and sorghum (*Sorghum bicolor*).

- **Maize generator**: Produces plants with heights ranging from 0.62 m to 1.44 m and leaf counts from 3 to 22. Morphological parameters are sampled from continuous Gaussian distributions coupled through four latent factors (vigor, slenderness, posture, and texture). This sampling leaves leaf count and plant height weakly correlated ($r = 0.20$).
- **Sorghum generator**: Uses the EvoEngine sorghum generator to produce plants with stem heights ranging from 0.19 m to 1.74 m and leaf counts from 4 to 35, with stem length strongly coupled to leaf count (stem length $= 0.05\, n_{\text{leaves}} - 0.001$, $R^2 = 0.988$).

Including both species enables us to evaluate whether our framework generalizes across both strongly coupled and weakly coupled morphological regimes.

Isolated-plant sensor rendering. For each generated plant instance, we render ten camera perspectives at 1024×1024 resolution (40° vertical field of view). For each perspective, we export a standard 2D RGB photograph, a metric depth frame stored losslessly as a 32-bit packed file, and camera calibration parameters. Sorghum captures use a fixed 10-camera Fibonacci sphere rig positioned at a distance of 2.5 m (spanning elevation angles from $-64°$ to $+64°$), and each view carries the occlusion-free whole-plant point cloud in the camera frame. Maize captures draw ten uniform random spherical perspectives with camera distances adapted to the plant's bounding box, generating partial 8,192-point clouds back-projected directly from depth images to mimic surface-only camera captures.

Physically simulated row neighbor occlusion. In agricultural fields, plants grow in dense rows where leaves physically touch and overlap. To model this occlusion, we construct simulated crop rows placing each target plant between two adjacent neighbors from the same dataset split at a realistic 0.15 m stem spacing. Rows are rendered across ten yaw orientation angles ($\theta \in \{0°, 10°, \ldots, 90°\}$) with uniform random neighbor orientation offsets ($\pm 15°$).

A physical contact solver resolves inter-plant leaf interpenetration while preserving natural shapes. Each leaf blade is modeled as an articulated chain of 8 cm segments connected by joints that bend along circular arcs. When two leaves collide, the solver applies exact triangle–triangle intersection checks and rotates joints by the minimum angle required to clear contact. Neighbors bend first; the main target plant bends only when neighbors reach their joint rotation limits.

A ground-rover camera captures full row scenes (yielding an RGB photograph, depth frame, and a 24,576-point scene cloud containing both target and neighbor geometry) alongside ground-truth target files containing the main target plant alone in its contact-bent shape (8,192 points). In maize rows, neighboring leaves account for 59–68% of the plant pixels in the frame and two thirds of the input point cloud; in sorghum rows, 23% of the target plant's pixels are hidden behind neighboring leaves on average across orientation angles.

Dataset partitioning. Plants are split into training (70%, 10,500 plants), validation (15%, 2,250 plants), and test sets (15%, 2,250 plants) by plant instance. Validation and test splits are enriched with extreme morphological outliers, selected via Mahalanobis distance ranking across core structural features.

3.3. Four-stream masked pretraining

Multimodal tokenization. Our network architecture builds on EmbodiedMAE [10], extending multi-sensor visual encoders to ingest structured procedural parameters alongside standard vision modalities:

1. **RGB and depth patches**: 2D RGB photographs and metric depth maps are partitioned into 196 non-overlapping patches on a 14×14 grid. Patches are linearly projected into 768-dimensional token embeddings added to 2D sine–cosine spatial positional embeddings.
2. **Point cloud groups**: Following Point-MAE [33], 3D point clouds are subsampled to 196 group centers using farthest point sampling (FPS). For each center, its 32 nearest spatial neighbors are grouped and processed by a lightweight PointNet encoder [36] to generate a 768-dimensional token.
3. **Procedural parameter tokens (θ)**: The parameter vector θ is formatted as $1 + K$ continuous numerical tokens: 1 token for global plant-level attributes and K tokens corresponding to individual leaf slots ($K = 28$ for maize, $K = 24$ for sorghum; plants with more leaves keep their first K slots). Rather than discretizing continuous numbers into string text, attributes are affinely normalized to $[0, 1]$ and projected linearly into embedding space as continuous numerical tokens via xVal-style encoding [15].

Learned 1D modality embeddings are added to distinguish streams, and a learnable `[CLS]` token is prepended to the input sequence. The sequence is processed by a 12-block ViT encoder (width 768) followed by an 8-block decoder (width 512). The decoder restores all L token positions with learnable mask tokens; linear heads predict RGB and depth patches, a FoldingNet head emits 41 points per point-cloud token, and an MLP regresses the continuous parameters. When predicted parameters are scored or re-rendered, the real leaf slots are selected by the ground-truth validity mask; inferring leaf presence from the sensors alone is left to future work.

Dynamic Dirichlet modality masking. To train a single network capable of completing any arbitrary subset of input modalities at test time, we apply a heavy overall token masking budget of $\rho = 0.80$. The total number of visible tokens V is given by:

$$V = \lfloor (1 - \rho)\, L \rfloor , \tag{2}$$

yielding $V = 122$ visible tokens for sorghum ($L = 613$ total tokens) and $V = 123$ for maize ($L = 617$).

At each pretraining step, V is allocated across input streams by sampling a proportion vector from a symmetric Dirichlet distribution, $\mathrm{Dirichlet}(\mathbf{1})$. The sampling is constrained such that every stream retains at least one visible token and at least 25% masked tokens. Because parameter tokens are included within this budget in the headline setup, the parameter stream remains $\sim$58% visible on average during pretraining.

To analyze parameter stream placement (Sec. 4.6), we also train control arms where parameter tokens are uncoupled from the sensor budget and masked at fixed visibility ratios ρ_θ:

$$V_\theta = \begin{cases} 0, & \rho_\theta = 1, \\ \max(1, \lfloor (1 + K)(1 - \rho_\theta) \rfloor), & 0 \le \rho_\theta < 1. \end{cases} \tag{3}$$

Setting $\rho_\theta = 1$ forces procedural parameters to act strictly as a decoder reconstruction target without entering the encoder.

Multi-term reconstruction objective. The pretraining objective optimizes four modality-specific reconstruction loss terms:

$$\mathcal{L}_{\mathrm{rec}} = \ell_{\mathrm{rgb}} + \ell_{\mathrm{d}} + \ell_{\mathrm{pc}} + 5\ell_\theta. \tag{4}$$

Here, ℓ_{rgb} is MSE computed on patch-normalized RGB pixels [21], ℓ_{d} is MSE on min–max normalized depth values, and ℓ_θ is Smooth-L_1 loss evaluated over masked parameter tokens.

For 3D point clouds, standard nearest-neighbor Chamfer distance allows multiple predicted points to match a single target point without penalty, encouraging decoders to collapse points into dense clumps on central stems while ignoring slender leaf blades [42, 47]. To address this limitation, we supervise point cloud prediction using a sigmoid-weighted Chamfer loss:

$$\ell_{\mathrm{pc}} = \frac{1}{|\hat{P}|} \sum_{p \in \hat{P}} w(\delta(p, P))\, \delta(p, P) + \frac{1}{|P|} \sum_{q \in P} w(\delta(q, \hat{P}))\, \delta(q, \hat{P}), \tag{5}$$

Dataset	Real scans	Slender	Row occl.	RGB+D+PC	Exact proc. GT	Scale
PCN / ShapeNet-55 [49, 51]	×	×	×	×	×	55,000
Pheno4D [41]	✓	✓	×	×	×	84
Crops3D [54]	✓	✓	×	×	×	2,500
MaizeField3D [25]	✓	✓	×	×	(✓)	1,000
FIELDPROC (ours)	✓†	✓	✓	✓	✓	30,000

Table 2. **Dataset comparison.** Exact Procedural GT indicates exact ground-truth generative parameter vectors per plant; (✓) denotes parameters fitted post hoc via optimization. †FIELDPROC includes real-field evaluation scans; training splits consist of procedural synthetic data.

where $\delta(p, P) = \min_{q \in P} \|p - q\|_2$, $w(r) = \sigma(a(r - \tau))$, with slope parameter $a = 100$ and spatial threshold $\tau = 0.01$ (relative to unit-sphere radius, $\approx$ 7 mm on a 1.4 m plant).

To further penalize missing leaf blades, our headline pretraining setup incorporates an optimal-transport term based on debiased Sinkhorn divergence S_ε [12]:

$$\ell^\star_{\text{pc}} = \ell_{\text{pc}} + 0.5\, S_\varepsilon\left(\hat{P}_{1024}, P_{1024}\right), \tag{6}$$

computed between random 1,024-point subsets of predicted and target point clouds ($\varepsilon = 0.02$). Sinkhorn divergence matches point sets as global probability distributions, forcing the decoder to cover the full spatial extent of slender leaf organs. The ablation arms of Sec. 4 use Eq. (5) alone; the Sinkhorn term enters the headline model and the distillation. Models are trained with AdamW ($\beta = (0.9, 0.95)$, weight decay 0.05, cosine decay): the headline schedule peaks at 6×10^{-4} over 1,000 epochs of all ten views at global batch 256, while ablation arms train for a matched budget of 197,400 optimizer steps at global batch 32, one view per plant per epoch, peaking at 1.5×10^{-4}.

3.4. Full-modality teacher to single-sensor students

While pretraining exposes the encoder to partial subsets of all four modalities, single-sensor field deployment (e.g., inputting only a single photograph) presents a configuration pretraining never shows the encoder: every token of the unobserved streams absent. To close this gap, we employ teacher–student feature distillation [22].

Distillation mechanism. The teacher model is a frozen four-stream network receiving all input modalities, with parameter tokens as privileged information: for sorghum every token of every stream is visible, while for maize the teacher retains half the tokens of each stream, since its point-cloud reconstruction degrades when fully visible. The teacher outputs a global encoder latent $c^T \in \mathbb{R}^{768}$ and normalized decoder feature maps $F^T \in \mathbb{R}^{L \times 512}$.

Student models are initialized from teacher weights and trained for 100 epochs using AdamW (learning rate 10^{-4}, global batch 128). During student training, 80% of steps perform single-source distillation where the student receives only a single source $s \in \{\text{rgb}, \text{d}, \text{pc}, \theta\}$, while 20% of steps perform standard four-stream pretraining to maintain feature stability. At deployment only sensor sources are available, and each is evaluated alone. The student is optimized via combined reconstruction and cross-modal feature alignment:

$$\mathcal{L}_{\text{student}} = \mathcal{L}_{\text{rec}} + \frac{\lambda_F}{|\mathcal{J}_{\neg s}|} \sum_{j \in \mathcal{J}_{\neg s}} \|F^S_j - F^T_j\|_2^2 + \lambda_c \|c^S - c^T\|_2^2, \tag{7}$$

where $\mathcal{J}_{\neg s}$ denotes decoder token positions belonging to unobserved non-source streams ($\lambda_F = 1.0, \lambda_c = 0.5$). Through Eq. (7), single-sensor student models learn to reconstruct complete 3D geometry and continuous procedural parameters guided by multi-sensor teacher latents.

3.5. Field evaluation and transfer protocols

We evaluate model performance across three operational regimes:

1. Phenotypic trait probing. To measure phenotypic trait information captured within encoder latents without relying on decoders, we fit linear ridge regression probes on frozen `[CLS]` tokens of the training plants and score them on the held-out validation and test plants, one deterministic view per plant. Procedural parameter inputs are zeroed during probing to prevent label leakage. Trait recovery is evaluated using the coefficient of determination (R^2) accompanied by 95% paired bootstrap confidence intervals (2,000 resamples over plants). A model modification is classified as beneficial only when its performance gain confidence interval excludes zero on both validation and test splits. Under partial point-cloud input, the probe is either refit per visibility condition (*matched*) or applied unchanged from the complete-cloud fit (*transferred*); partial-cloud reconstruction keeps the masked input regime of pretraining.

2. Row occlusion protocol. Models are evaluated on simulated crop rows under two configurations:

- **Zero-shot transfer**: Feeding row observations directly into isolated-trained models to evaluate sensitivity to neighbor occlusion.
- **Row fine-tuning**: Training decoders to reconstruct target main plants behind occluding neighbors while inputting full row scene observations.

A control arm runs the same fine-tuning schedule on the isolated twins, and generated clouds are scored against the target plant and against the input scene, so a model that merely copies the scene is caught.

3. Real-field validation. Real-field performance is evaluated on field captures from the MaizeField3D dataset [25]. Predicted procedural parameters and re-rendered 3D meshes are compared against the procedural fits of a staged vision-language pipeline [18], which serve as independently reconstructed ground truth.

4. Experiments

In this section, we present an extensive experimental evaluation of CROPCODE. We describe our experimental setup (Sec. 4.1), followed by evaluations across three deployment regimes: isolated single-view reconstruction (Sec. 4.2), difficulty progression across viewpoint and self-occlusion (Sec. 4.3), and neighbor occlusion in realistic field rows (Sec. 4.4). We then analyze cross-modal distillation (Sec. 4.5), investigate the placement of the procedural parameter stream (Sec. 4.6), uncover systematic divergences between geometric metrics and downstream trait probes (Sec. 4.7), describe the real-field validation protocol (Sec. 4.8), and analyze scaling, cross-species transfer, and latent space structure (Sec. 4.9).

4.1. Experimental setup

Models and compute budgets. To establish controlled comparisons across ablations, all variant models share a standardized step budget of 197,400 optimizer updates at a global batch size of 32 (corresponding to 600 epochs sampling one random camera view per plant over the 10,500 training plants). All controlled ablation variants use identical view-sampling random seeds and a ViT-Base encoder (114 M parameters) unless model scale is explicitly varied. Experimental arms that vary the volume of training plants adjust epoch counts proportionally to maintain an identical step budget (Sec. 3.3).

The headline full-modality model is trained on an extended schedule of 1,000 epochs across all ten camera views at a global batch size of 256, incorporating the Sinkhorn optimal-transport point cloud loss (Sec. 3.3). The headline model encodes the procedural parameter stream within its shared Dirichlet masking budget and serves as the weight initialization for all teacher and student models in Sec. 4.5. Results are reported on the validation split unless stated otherwise.

Evaluation metrics. We evaluate 3D geometric accuracy using symmetric Chamfer distance on squared point displacements in the unit-sphere frame ($\times 10^{-3}$; a value of 1.0 corresponds to a root-mean-square displacement of $\approx$ 22 mm on a 1.4 m plant). For pretrained models, Chamfer distance is evaluated in the masked input regime of pretraining; for distilled student models, it is evaluated with unmasked sensor inputs.

Parameter regression accuracy is quantified via mean absolute error (MAE) evaluated over normalized $[0, 1]$ continuous fields of hidden, non-padding leaf parameter tokens. Where per-trait relative performance is informative, we report trait *skill*, defined as

$$\text{Skill} = 1 - \frac{\text{MAE}}{\text{MAE}_{\text{mean}}}, \tag{8}$$

where 1.0 indicates perfect parameter recovery, 0.0 represents performance equivalent to predicting the dataset mean, and negative values indicate predictions worse than the mean baseline.

Phenotypic trait representation quality is evaluated with linear ridge regression probes fit on frozen encoder `[CLS]` tokens with all procedural parameter inputs zeroed to prevent label leakage. Probes are fit on a single deterministic view per training plant and evaluated on held-out validation and test sets (2,250 validation, 2,250 test plants). Performance is reported as coefficient of determination (R^2) accompanied by 95% paired bootstrap confidence intervals (2,000 resamples over plants). A model modification is classified as a *win* only when its gain's confidence interval excludes zero on both validation and test splits simultaneously.

Species developmental traits. In sorghum, four core developmental traits have effective rank two due to generator allometry (height and leaf count are tightly coupled, $r = 0.994$). We therefore supplement sorghum trait probing with independent parameter axes varied by the generator, including leaf roll and leaf blade length. In maize, eleven largely independent morphological traits are evaluated directly (Sec. 3.2).

4.2. Regime I: single-view reconstruction of isolated plants

Table 3 presents performance results for single-view deployment on isolated crop plants. Given a single photograph, a single depth frame, or a partial point cloud, CROPCODE predicts the full procedural parameter vector and completes unobserved sensor modalities in a single feed-forward pass.

Input Modality	Leaf Count (Acc. ± 1) ↑	Height Error (Rel. Err.) ↓	Leaf Parameters (Mean Skill) ↑	F_1-score ($\tau = 0.01$) ↑	Chamfer $\times 10^{-3}$ ↓
Sorghum					
RGB (1 view)			0.69*		0.88
RGB (3 views)					
Depth (1 view)					0.62
Point Cloud (1 view)			0.76*		1.22
Full Teacher (RGB + D + PC)					**0.35**
Two-Stage Baseline (Completion → Fit)					
Maize					
RGB (1 view)					0.77
RGB (3 views)					
Depth (1 view)					0.58
Point Cloud (1 view)					0.83
Full Teacher (RGB + D + PC)					
Two-Stage Baseline (Completion → Fit)					

Table 3. **Regime I: single-view reconstruction on isolated plants.** Each row reports performance for the distilled generalist model given only the specified sensor input, compared against the full-modality teacher model and a two-stage baseline (point cloud completion followed by parameter fitting). Chamfer distance ($\times 10^{-3}$, validation) is evaluated against complete ground-truth point clouds. *Mean skill across active leaf parameter fields (insertion height, branching angle, length, and roll). Leaf-count accuracy, height error, F_1, the 3-view and two-stage rows, and the maize teacher are not yet evaluated; the empty cells await the FIELDPROC evaluation scripts.

On sorghum, point clouds generated from a single photograph achieve a Chamfer distance of 0.88×10^{-3} (0.79×10^{-3} on test), single depth frames yield 0.62×10^{-3}, and point cloud regeneration achieves 1.22×10^{-3}, compared to 0.35×10^{-3} for the full-modality teacher. The photograph-to-cloud student achieves an overall leaf parameter recovery skill of 0.984 relative to split variance. Evaluating active leaf parameter fields, an input point cloud achieves skill scores of 0.95 for leaf insertion height, 0.93 for branching angle, 0.67 for leaf length, and 0.50 for leaf roll. A single photograph achieves skill scores of 0.90, 0.85, 0.66, and 0.33 on the same fields, respectively. Point clouds outperform single photographs on every field exhibiting nonzero skill, with the performance gap largest on leaf roll.

Figure 3 illustrates qualitative single-photograph reconstructions on held-out test plants. While the headline model before distillation generates diffuse point clouds around target silhouettes, the distilled generalist model resolves individual leaf blades, accurately recovering normalized leaf length profiles along the central stem from base to apex. Furthermore, because the output representation is procedural, predicted parameters can be interactively modified (such as adjusting individual blade insertion angles) and re-rendered to create editable, simulation-ready 3D models.

4.3. Regime II: viewpoint elevation, self-occlusion, and complexity

Viewpoint elevation. Camera perspectives form an elevation ladder spanning angles from steep top-down to steep bottom-up ($\sin \text{elev} = -0.9 + 0.2\,i$). Because every perspective of a plant shares identical 3D ground-truth geometry and procedural parameters, varying the viewpoint alters sensor input while keeping target representations constant.

Figure 4A plots photograph-to-cloud Chamfer error across camera elevation angles. Reconstruction error follows a symmetric U-shaped distribution: error is lowest for horizontal side-on perspectives and highest for views looking steeply downward or upward. Distillation reduces reconstruction error by 28% on average across views while preserving the overall U-shaped difficulty curve. Across 2,250 test plants, camera elevation accounts for $\eta^2 = 0.24$–0.31 of per-plant reconstruction error variance. In contrast, plant morphological traits (leaf count, stem height, slenderness ratio) exhibit negligible correlation with reconstruction error ($|\rho| \leq 0.06$). Thus, camera perspective rather than plant morphology dominates single-image inversion difficulty.

Training view distribution. We pretrain models under restricted camera elevation bands: top-down views ($i \in \{0, 1\}$), horizontal views ($i \in \{4, 5\}$), or a single top-down view ($i = 0$), benchmarked against pretraining across all ten views. Probes are evaluated across held-out views k (Fig. 5). Restricted models achieve superior performance on their specific training elevations (e.g., $+0.061$ roll R^2 on view 0 for the single-view model), but experience performance drops on unobserved elevations (falling by -0.066 to -0.144 on distant views). Models trained on the complete elevation ladder maintain stable performance across all perspective angles ($R^2 = 0.734$–0.778).

Self-occlusion under partial camera coverage. To evaluate model performance under partial observations, point clouds are clipped via mesh ray-casting to simulate surface points visible from 1 or 3 camera perspectives (observing 49% and 78% of total plant surface area, respectively; Tab. 4).

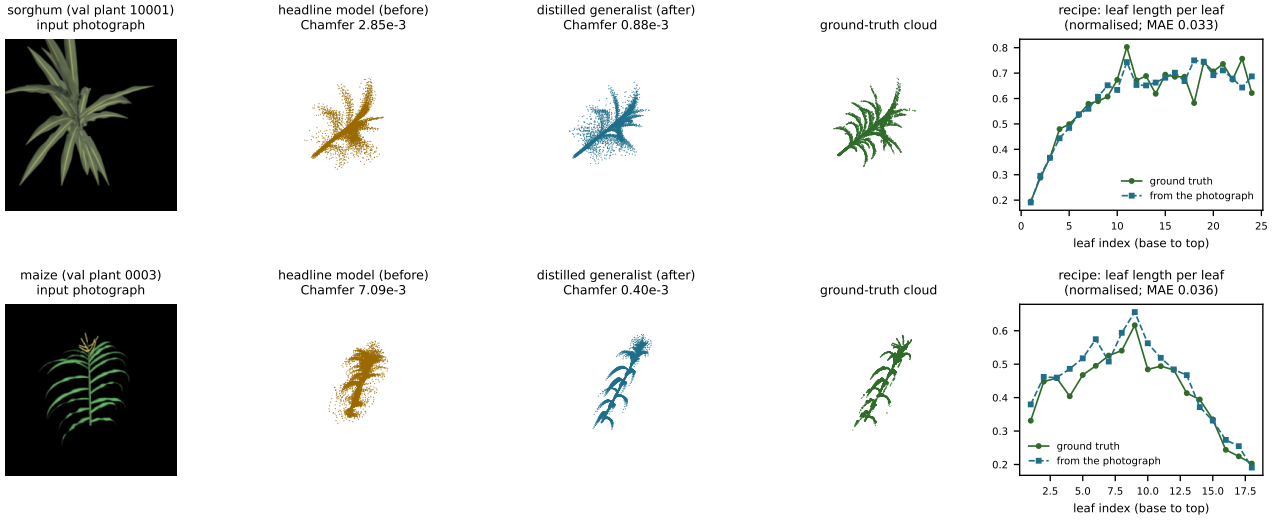


Figure 3. **Single-photograph 3D plant reconstruction on held-out test plants.** Each row shows inference given only a single input photograph (left). The headline model before distillation and the distilled generalist generate 3D point clouds, with Chamfer distance indicated above. The rightmost panel demonstrates parameter recovery by comparing the generalist's recovered normalized leaf lengths with exact ground truth across leaf indices from base to top.

	Matched Probe (R^2, Height)			Transferred Probe (R^2, Height)			Chamfer ($\times 10^{-3}$)		
Input Modality Cloud Coverage	Full 100%	3-Cam 78%	1-Cam 49%	Full 100%	3-Cam 78%	1-Cam 49%	Full 100%	3-Cam 78%	1-Cam 49%
Point Cloud (PC)	0.888	0.839	0.765	0.888	0.509	0.575	4.99	6.18	12.8
PC + RGB	0.977	0.967	0.948	0.977	0.624	0.001	2.49	3.64	11.5
PC + RGB + D	0.975	0.968	0.956	0.975	0.721	0.576	1.76	3.06	11.4
Four-Stream Model	0.961	0.953	0.946	0.961	0.748	0.818	2.52	3.62	11.4
Headline Teacher	**0.982**	**0.975**	**0.960**	**0.982**	0.623	−0.26	**1.36**	3.14	11.6

Table 4. **Self-occlusion under partial camera coverage (sorghum test).** Evaluated on point clouds restricted to surface points visible from 1 or 3 camera perspectives via mesh ray-casting. Matched: linear probe refit per visibility condition. Transferred: linear probe trained on complete clouds applied directly without fine-tuning. Chamfer distance is evaluated against the complete ground-truth plant point cloud.

Refitting linear probes under each visibility condition demonstrates that phenotypic trait information persists within encoder latents despite partial coverage: height R^2 remains high at 0.960 for the headline model under 1-camera coverage. However, transferring probes trained on full point clouds directly to partial coverage inputs results in catastrophic failure ($R^2 \to -0.26$, Fig. 6). This reveals that partial visibility induces a covariate shift in latent feature space, even though underlying morphological trait information remains linearly decodable when probes are refit.

Morphological complexity. Reconstruction performance remains consistent across morphological scale variations. Figure 7 plots photograph-to-cloud Chamfer distance across leaf-count deciles for sorghum and maize. Following teacher–student distillation, reconstruction error is uniform across leaf-count deciles (Spearman $\rho = -0.07$ for sorghum, $+0.04$ for maize), confirming that model performance does not degrade as organ count increases.

4.4. Regime III: neighbor occlusion in realistic crop rows

Table 5 compares performance on identical target plants rendered in isolation versus within simulated crop rows flanked by adjacent neighbors. Models trained exclusively on isolated plants experience severe performance degradation under row occlusion, increasing Chamfer error sixteenfold on maize ($1.8 \to 29.0 \times 10^{-4}\,\mathrm{m}^2$) and tenfold on sorghum ($13.1 \to 133.8 \times 10^{-4}\,\mathrm{m}^2$).

Evaluating models on the "echo test" (comparing Chamfer distance against input scene clouds versus target-only clouds) reveals that isolated control models inadvertently reconstruct neighboring foliage as belonging to the target plant. Row fine-tuning (training decoders to predict target main plant geometry behind occluding neighbors) eliminates 92% of neighbor-induced reconstruction error

Model Variant	Sensor Input	Chamfer vs. Target (10^{-4} m^2) Isolated	Row	Δ [95% CI]	Leaf Parameter MAE Isolated	Row	Δ [95% CI]
Maize: Zero-Shot (Headline Model)							
	RGB	56.7	59.7	$+3.0$ [$+2.4$, $+3.6$]	0.085	0.088	$+0.003$ [$+0.003$, $+0.003$]
	Depth	47.2	62.6	$+15.3$ [$+14.8$, $+15.9$]	0.090	0.093	$+0.002$ [$+0.002$, $+0.003$]
	Point Cloud	20.0	33.8	$+13.7$ [$+13.4$, $+14.1$]	0.064	0.081	$+0.018$ [$+0.017$, $+0.018$]
	RGB + D + PC	26.4	29.6	$+3.2$ [$+2.9$, $+3.6$]	0.086	0.093	$+0.007$ [$+0.007$, $+0.008$]
Maize: Row Fine-Tuned							
	RGB	13.7	14.4	$+0.6$ [$+0.4$, $+0.8$]	0.067	0.071	$+0.004$ [$+0.003$, $+0.004$]
	Depth	10.3	9.6	-0.7 [-0.9, -0.5]	0.061	0.064	$+0.003$ [$+0.003$, $+0.004$]
	Point Cloud	21.7	21.9	$+0.2$ [-0.1, $+0.5$]	0.063	0.073	$+0.010$ [$+0.010$, $+0.010$]
	RGB + D + PC	12.6	**4.0**	-8.6 [-8.8, -8.3]	**0.046**	**0.052**	$+0.005$ [$+0.005$, $+0.005$]
Maize: Isolated Control							
	RGB	11.6	56.1	$+44.5$ [$+44.0$, $+45.1$]	0.071	0.095	$+0.024$ [$+0.023$, $+0.024$]
	Depth	4.6	52.8	$+48.2$ [$+47.5$, $+48.9$]	0.059	0.097	$+0.038$ [$+0.037$, $+0.039$]
	Point Cloud	8.2	25.1	$+16.9$ [$+16.6$, $+17.3$]	0.046	0.076	$+0.030$ [$+0.029$, $+0.030$]
	RGB + D + PC	1.8	29.0	$+27.2$ [$+26.9$, $+27.6$]	0.043	0.093	$+0.050$ [$+0.049$, $+0.051$]
Sorghum: Zero-Shot (Headline Model)							
	RGB	163.1	218.5	$+55.4$ [$+52.6$, $+58.4$]	0.129	0.135	$+0.006$ [$+0.006$, $+0.006$]
	Depth	106.2	207.2	$+101.0$ [$+98.0$, $+104.2$]	0.070	0.109	$+0.039$ [$+0.039$, $+0.040$]
	Point Cloud	177.7	231.4	$+53.8$ [$+52.2$, $+55.4$]	0.114	0.127	$+0.013$ [$+0.012$, $+0.013$]
	RGB + D + PC	174.3	230.5	$+56.2$ [$+53.9$, $+58.5$]	0.108	0.114	$+0.006$ [$+0.006$, $+0.006$]
Sorghum: Row Fine-Tuned							
	RGB	34.8	29.8	-5.0 [-5.4, -4.7]	0.094	0.097	$+0.003$ [$+0.003$, $+0.004$]
	Depth	24.4	18.6	-5.8 [-6.1, -5.5]	0.078	0.055	-0.022 [-0.023, -0.022]
	Point Cloud	46.0	70.8	$+24.8$ [$+24.4$, $+25.2$]	0.074	0.082	$+0.007$ [$+0.007$, $+0.007$]
	RGB + D + PC	38.7	**30.7**	-8.0 [-8.5, -7.6]	**0.073**	**0.061**	-0.012 [-0.013, -0.011]
Sorghum: Isolated Control							
	RGB	15.7	133.2	$+117.5$ [$+115.8$, $+119.3$]	0.094	0.114	$+0.020$ [$+0.020$, $+0.021$]
	Depth	6.6	171.9	$+165.3$ [$+162.9$, $+167.8$]	0.042	0.100	$+0.059$ [$+0.058$, $+0.060$]
	Point Cloud	30.9	100.9	$+70.0$ [$+68.8$, $+71.3$]	0.065	0.083	$+0.018$ [$+0.018$, $+0.019$]
	RGB + D + PC	13.1	133.8	$+120.8$ [$+119.5$, $+122.1$]	0.049	0.096	$+0.047$ [$+0.046$, $+0.048$]

Table 5. **Regime III: comparative performance of identical plants in isolation versus crop rows.** Evaluated across 2,250 held-out plants per species at 10 row orientations ($N = 22{,}500$). Chamfer distance is reported in metric units (10^{-4} m^2). Δ indicates the paired within-plant difference (row minus isolated) with 95% bootstrap confidence intervals over plants.

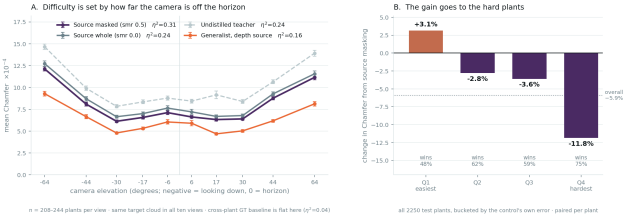


Figure 4. **Viewpoint elevation determines reconstruction difficulty.** (A) Photograph-to-cloud Chamfer distance of test plants plotted by camera elevation angle (208–244 plants per elevation view). Pretrained and distilled single-sensor models follow a symmetric U-shaped error curve centered on horizontal side views; camera elevation accounts for $\eta^2 = 0.24$–0.31 of per-plant reconstruction variance. (B) Performance gain from masking 50% of source tokens during distillation, grouped by error quartile of the control model, demonstrating concentrated improvement on challenging plant geometries.

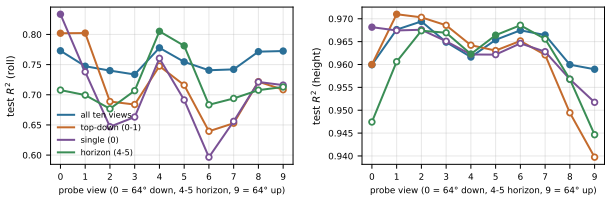


Figure 5. **Effect of training view distribution on downstream phenotype probing (sorghum test).** Probes are fitted and evaluated on individual views k for model variants trained on all ten views versus restricted bands of the elevation ladder. Filled markers indicate training views. Restricted training view bands improve probe performance specifically within those elevations but degrade generalization on unobserved view angles.

in maize ($29.0 \rightarrow 4.0 \times 10^{-4}$ m^2) and 85% in sorghum ($133.8 \rightarrow 30.7 \times 10^{-4}$ m^2).

Figure 8 shows qualitative reconstructions under row occlusion. While isolated control models generate spurious leaf structures corresponding to neighboring plants, row-fine-tuned models isolate target main plant geometry behind occluding foliage. Figure 9 plots reconstruction error across row orientation angles (θ). In maize rows, occlusion error peaks broadside ($\theta = 0°$, where leaf planes align along the row) and drops end-on ($\theta = 90°$). In sorghum rows, plant headings are drawn uniformly at random, so occlusion error remains constant across all yaw angles.

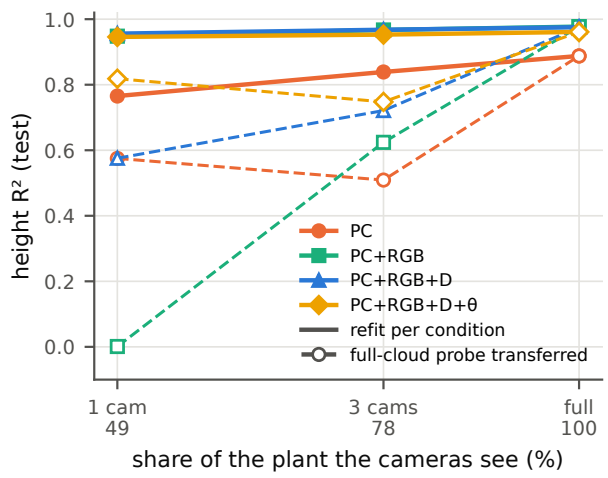


Figure 6. **Phenotypic information persistence under partial visibility.** Plant height prediction performance (R^2, test set) plotted against the visible percentage of the plant under 1-camera, 3-camera, and full coverage. Solid lines represent probes refit per visibility condition; dashed lines show direct transfer of the full-cloud readout probe.

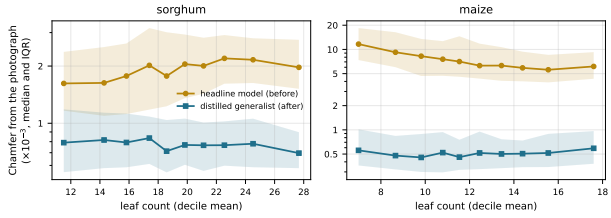


Figure 7. **Error against plant complexity (test, 2,250 plants per species).** Chamfer distance of point clouds generated from single photographs across leaf-count deciles for the headline model before distillation and the distilled generalist.

4.5. Any-subset modality reconstruction and distillation

Table 6 reports cross-modal generation performance before and after teacher–student feature distillation. Distillation from the full-modality teacher reduces mean cross-modal generation loss by 30.2% in sorghum and 43.5% in maize.

Reconstruction from single photographs improves substantially: on sorghum, photograph-to-cloud Chamfer error drops by 61% ($2.27 \rightarrow 0.88 \times 10^{-3}$) and parameter MAE falls by 76% ($0.058 \rightarrow 0.014$). On point-cloud inputs, parameter regression MAE is halved ($0.040 \rightarrow 0.020$). Dual-stream inputs combining sensor data with procedural parameters (RGB + θ and D + θ) achieve the overall lowest point cloud reconstruction errors (0.36–0.44×10^{-3}).

4.6. Influence of procedural parameter stream placement

Tables 7 and 8 analyze stream contributions and parameter token placement.

Modality value-add. Visual image streams provide the primary source of trait representation quality. On sorghum, adding RGB or depth to point clouds increases height R^2 from 0.895 to 0.979 (RGB) and 0.976 (depth). On maize, adding RGB increases height R^2 from 0.171 to 0.620. Adding depth to PC + RGB reduces Chamfer distance by 28% in sorghum ($2.57 \rightarrow 1.85 \times 10^{-3}$) and 25% in maize ($1.61 \rightarrow 1.21 \times 10^{-3}$), but provides negligible improvement on downstream linear probe traits.

Parameter stream placement dynamics. A critical architectural question is where privileged procedural parameter tokens should enter a multimodal masked autoencoder. When parameter tokens are visible to the encoder during pretraining, they state the targets outright; one interpretation is that the network relies on these parameter tokens as an informational shortcut rather than learning visual features from sensor patches, so that when parameter tokens are absent at deployment (where only sensor streams exist), sensor representations are impoverished.

In budget-matched controls (Tab. 8), restricting parameter visibility to 0% (target-only) or 20% reveals species-specific dynamics:

- **Sorghum**: Target-only parameter supervision degrades sensor latent probing quality (R^2 falls to 0.966). Sorghum parameter attributes are tightly coupled (effective rank two); a plausible mechanism is gradient competition during multi-task decoding that suppresses visual feature learning.
- **Maize**: Target-only parameter supervision improves trait probe scores across all eleven targets ($R^2 = 0.776$ vs. 0.633 on height). Maize parameters span eleven largely independent dimensions; a plausible mechanism is that structured multi-task decoder gradients regularize sensor feature representations.

4.7. Divergence between Chamfer distance and probing metrics

Figure 12 compares validation Chamfer distance against frozen-latent probe R^2 across twelve model variants. While the two metrics show overall rank correlation (Spearman $\rho = -0.57$– -0.88), they diverge systematically on key design choices:

1. **Depth inclusion**: Adding depth maps reduces Chamfer distance by 25–28% while leaving linear probe trait recovery unchanged.

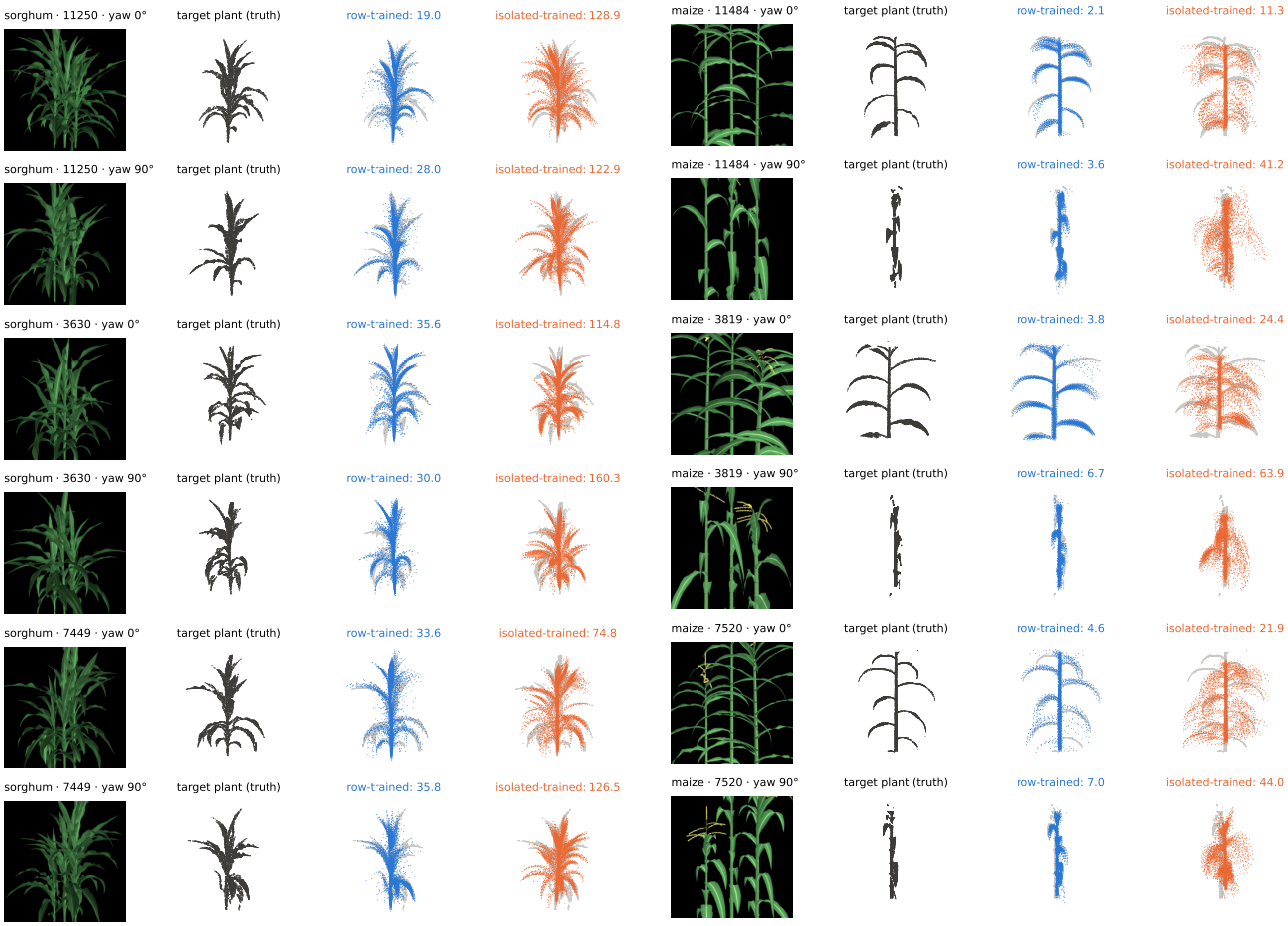


Figure 8. **Row occlusion reconstruction on held-out plants.** Sorghum (left) and maize (right) plants shown broadside (0°) and end-on (90°). Columns display raw row photographs, ground-truth target plant geometry, and predicted point clouds from row-trained models (blue) versus isolated-trained controls (orange).

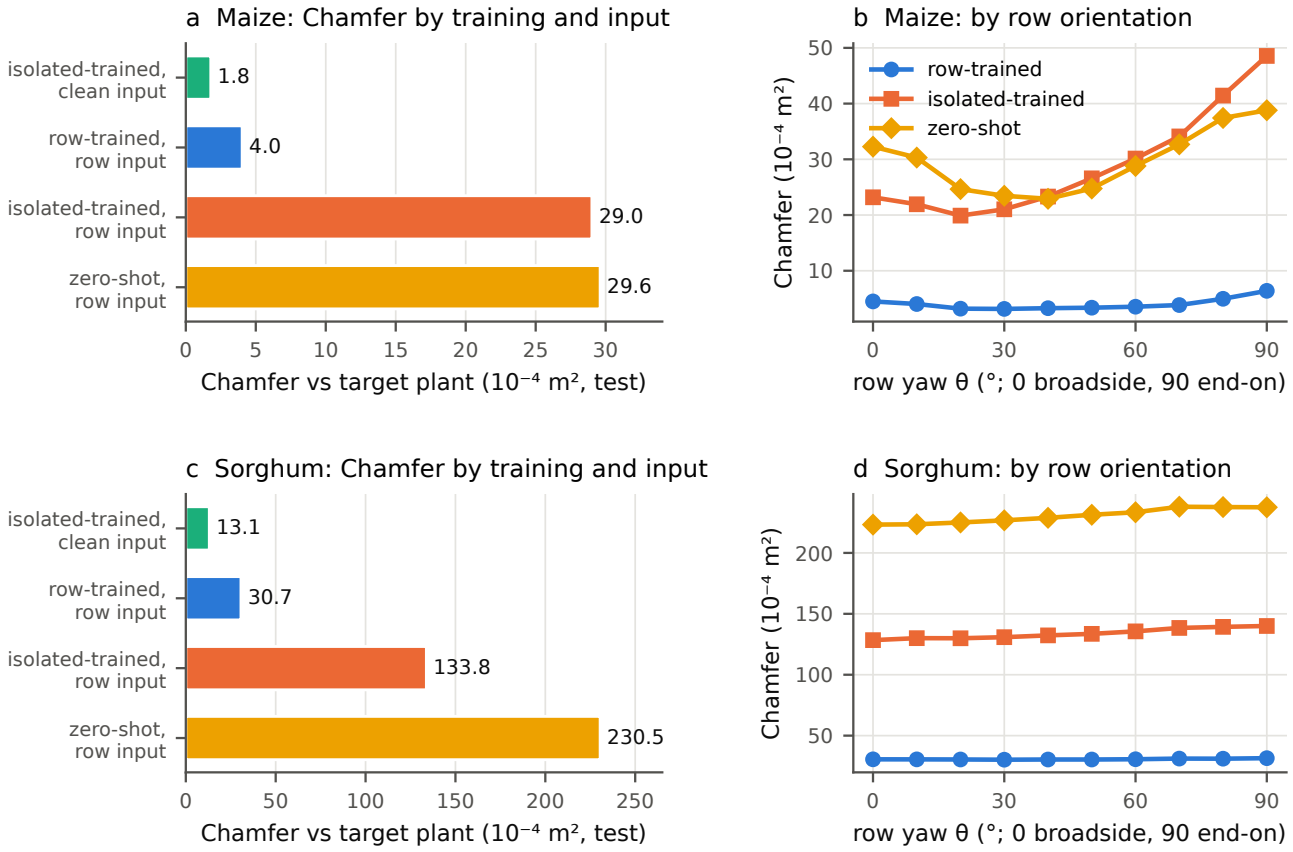


Figure 9. **Quantitative row occlusion performance on maize (a, b) and sorghum (c, d) test rows.** (a, c) Target plant Chamfer distance comparing isolated-trained models on clean versus row inputs, row-trained models on row inputs, and zero-shot baselines. (b, d) Reconstruction error plotted across row orientation yaw angles (θ). Fine-tuning on row scenes eliminates the vast majority of neighbor-induced occlusion error in both species.

	Sorghum (Validation Set)				Maize (Validation Set)			
	θ MAE ↓		Chamfer $\times 10^{-3}$ ↓		θ MAE ↓		Chamfer $\times 10^{-3}$ ↓	
Input Modality	Before	After	Before	After	Before	After	Before	After
RGB Only	0.058	0.014	2.27	0.88	0.037	0.019	11.3	0.77
Depth Only	0.017	**0.013**	0.90	0.63	0.024	0.019	1.26	0.58
Point Cloud Only	0.040	0.020	5.48	1.39	0.020	0.018	6.43	0.83
θ (Parameters Only)	–	–	22.7	6.83	–	–	80.7	12.8
Two-Stream Input Pairs → Point Cloud Generation (Dedicated Student / Generalist)								
RGB + D	–	–	2.91	0.60 / 0.59	–	–	1.30	0.54 / 0.64
RGB + θ	–	–	0.43	0.40 / 0.39	–	–	0.47	0.44 / 0.43
D + θ	–	–	0.50	0.37 / **0.36**	–	–	0.46	0.42 / 0.40
Mean Generation Loss	0.371 → 0.259 (−30.2%)				0.290 → 0.164 (−43.5%)			

Table 6. **Any-subset modality reconstruction before and after cross-modal distillation (validation set).** Each row represents evaluation given only the specified source modality to generate remaining streams. Before: zero-shot pre-trained model warm-start; After: distilled generalist student model. Maize RGB+D (epoch 65) and three-pair (epoch 35) students still training: interim values.

2. **Parameter visibility**: Parameter tokens degrade sensor latent probing performance in sorghum while improving surface Chamfer distance in maize.

Chamfer distance evaluates nearest-neighbor surface proximity ($\min_q \|p-q\|_2$). On slender crop geometry, placing points near central stems minimizes Chamfer distance even if entire leaf blades are omitted. In contrast, downstream linear probing evaluates structural phenotype recovery. The ablation of the Sinkhorn term (E7, run by collaborators) is pending; its effect on Chamfer and F_1 will be

Stream Set	Chamfer $\times 10^{-3}$	Frozen-Probe R^2 (Validation Set) Height	Leaf Cnt	Biomass	Branch	Roll	Leaf Len
Sorghum							
PC	5.00	0.895	0.896	0.892	0.792	0.393	0.088
PC + D	2.45	0.976	0.979	0.982	0.903	0.798	0.307
PC + RGB	2.57	**0.979**	**0.983**	**0.986**	**0.934**	0.808	0.267
PC + RGB + D	**1.85**	0.977	0.980	0.982	0.926	0.781	**0.309**
PC + θ	4.84	0.860	0.864	0.863	0.780	0.481	0.078
PC + D + θ	4.08	0.907	0.909	0.916	0.856	0.694	0.191
PC + RGB + θ	3.00	0.962	0.964	0.967	0.890	**0.810**	0.225
PC + RGB + D + θ	2.53	0.962	0.964	0.965	0.880	0.760	0.249
		Height	Leaf Cnt	Biomass	Branch	Curl	Stem Rad
Maize							
PC	2.63	0.171	0.670	0.610	0.756	0.114	0.289
PC + D	2.55	0.453	0.773	0.683	0.804	0.082	0.550
PC + RGB	1.61	0.620	**0.914**	**0.783**	**0.927**	0.334	0.753
PC + RGB + D	1.21	0.633	**0.914**	0.774	0.926	0.236	0.760
PC + θ	2.49	0.256	0.740	0.662	0.787	0.114	0.363
PC + D + θ	2.54	0.382	0.801	0.670	0.795	0.136	0.408
PC + RGB + θ	1.21	0.509	0.881	0.780	0.912	**0.371**	**0.763**
PC + RGB + D + θ	**0.99**	**0.635**	0.908	0.781	0.919	0.304	0.743

Table 7. **Modality value-add ablations (E2).** Pre-trained models trained at equal optimizer step budgets differing only in active input token streams. θ indicates inclusion of the parameter stream as an encoder input under the shared Dirichlet budget.

Variant	Vision Budget	θ Vis. Ratio	Chamfer $\times 10^{-3}$	R^2 (Validation Set) Height	Leaf Cnt	Roll	Leaf Len
Sorghum							
PC + RGB + D	117.0	–	**1.85**	**0.977**	**0.980**	**0.781**	**0.309**
+θ, Shared Budget	107.4	58%	2.53	0.962	0.964	0.760	0.249
+θ, Gated 20%	117.0	20%	2.29	0.964	0.967	0.771	0.278
+θ, Gated 60%	117.0	60%	2.40	0.959	0.963	0.761	0.239
+θ, Target Only	117.0	0%	2.49	0.966	0.970	0.702	0.220
Maize							
PC + RGB + D	117.0	–	1.21	0.633	0.914	0.236	0.760
+θ, Shared Budget	107.4	58%	0.99	0.635	0.908	0.304	0.743
+θ, Gated 20%	117.0	20%	0.92	0.704	0.931	**0.464**	0.791
+θ, Gated 60%	117.0	60%	1.05	0.521	0.851	0.303	0.688
+θ, Target Only	117.0	0%	**0.88**	**0.776**	**0.947**	0.357	**0.804**

Table 8. **Token-budget control ablations.** Four-stream models re-trained with parameter tokens (θ) placed outside the shared Dirichlet budget, maintaining an identical 117.0 visible sensor token allocation across parameter visibility levels (0%, 20%, 60%, and target-only).

reported here.

Figure 13 plots reconstruction precision, recall, and F_1-score across spatial thresholds τ. Across all variants, precision exceeds recall ($P > R$), confirming that nearest-neighbor Chamfer loss rewards decoders that cluster points onto central stem structures while omitting thin outer leaf extremities.

4.8. Real-world field validation

We validate CROPCODE on real field laser scans and RGB images from the MaizeField3D benchmark [25], comparing single-forward-pass predictions against ground-truth procedural models reconstructed by a staged vision-language annotation pipeline [18]. E10 has not yet run. This subsection's numbers (parameter accuracy against the pipeline fits, re-rendered Chamfer against the laser scans, zero-shot and fine-tuned rows, runtime per plant) are pending the real-scan evaluation.

4.9. Representation quality, scaling, and transfer

Comparison against published encoders. Table 9 compares CROPCODE against self-supervised visual encoders. Three-stream CROPCODE (PC + RGB + D) outperforms DINOv2, MultiMAE, Point-MAE, and EmbodiedMAE across 8 of 9 sorghum traits and 10 of 11 maize traits.

Camera pose artifact analysis. Evaluating Point-MAE with full camera pose matrices yields strong maize performance. However, this is largely a synthetic renderer artifact: synthetic maize renders place leaf planes along a canonical world plane, so camera pose matrices leak plant azimuth orientation. Evaluating Point-MAE with gravity-only camera poses (realistic field calibration) reduces performance (R^2 drops from 0.720 to 0.588 on height), while CROPCODE performance remains unaffected.

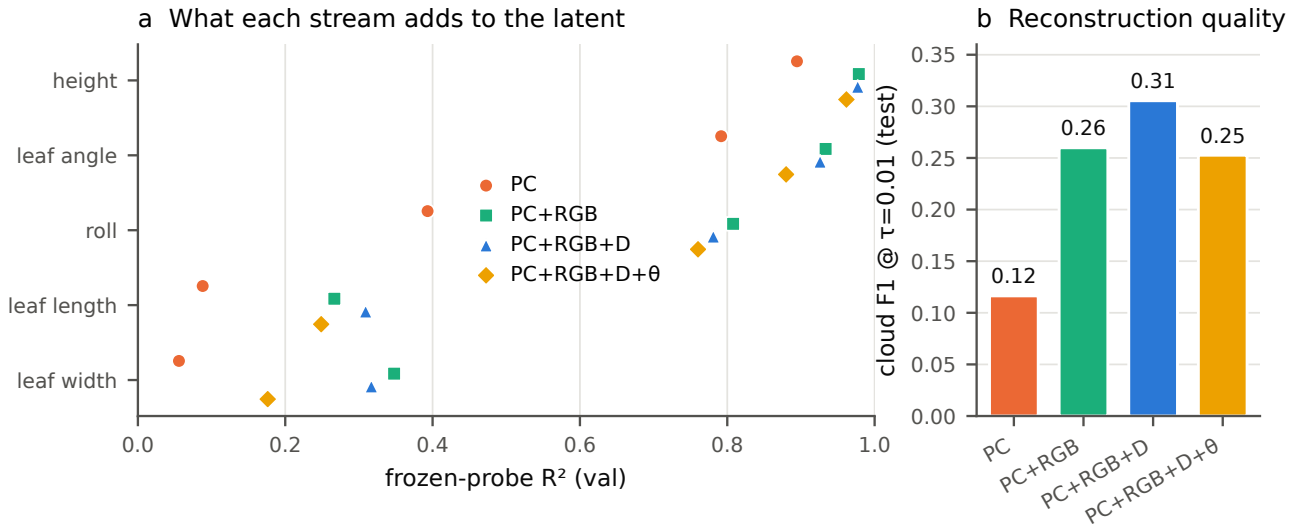


Figure 10. **Modality contributions to latent representation and geometry reconstruction (sorghum).** (a) Frozen-probe R^2 across phenotypic traits. (b) Reconstruction F_1 score at threshold $\tau = 0.01$ (≈ 1 cm).

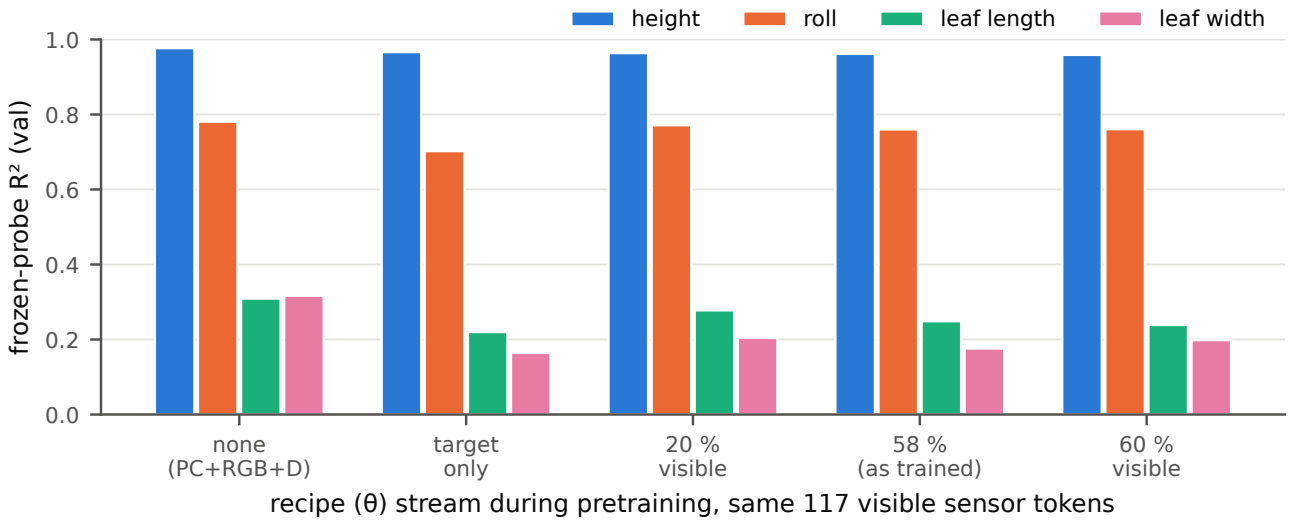


Figure 11. **Influence of procedural parameter stream visibility during pretraining.** Linear probe R^2 performance across parameter visibility levels under matched sensor token budgets.

Data volume and encoder capacity scaling. Table 10 and Figure 14 evaluate dataset size and network capacity at equal step budgets (197,400 steps). Scaling dataset volume improves Chamfer distance monotonically (sorghum $3.13 \rightarrow 2.53 \times 10^{-3}$; maize $3.54 \rightarrow 0.99 \times 10^{-3}$), while probe scores are not monotone in maize (height $0.393 \rightarrow 0.635$ with dips at intermediate scales). Scaling model capacity from ViT-Base (114M) to ViT-Large (332M) degrades representation quality under fixed compute steps.

Cross-species latent transfer. Pretraining on one crop species and fine-tuning on another for 150 epochs accelerates convergence and improves trait probing across all targets compared to training from scratch (sorghum roll R^2 increases by $+0.18$; maize blade curl increases by $+0.49$). However, while encoders transfer structural features cross-species, decoders fail to transfer zero-shot, yielding high Chamfer reconstruction errors on cross-species inputs.

Latent space visualization. Figure 15 visualizes t-SNE

Frozen Encoder Baseline	Sorghum					Maize			
	Height	Leaf Cnt	Branch	Roll	Leaf Len	Height	Branch	Curl	Stem Rad
DINOv2 ViT-B/14 (RGB) [32]	0.924	0.928	0.840	0.573	0.235	0.561	0.816	0.072	0.588
MultiMAE-B (RGB+D) [3]	0.922	0.925	0.833	0.679	0.232	0.570	0.783	0.069	0.442
Point-MAE (PC) [33]	0.958	0.963	0.863	0.716	0.188	0.509	0.863	0.199	0.426
EmbodiedMAE-B (RGB+D+PC) [10]	0.976	0.979	0.871	0.765	0.265	0.528	0.823	0.139	0.428
Supervised from Scratch (Ours)	0.973	0.978	**0.924**	0.634	0.271	**0.785**	**0.935**	**0.618**	**0.860**
CROPCODE PC Only	0.906	0.909	0.801	0.444	0.107	0.171	0.756	0.114	0.289
CROPCODE PC + RGB + D	**0.979**	**0.983**	0.903	**0.809**	**0.337**	0.633	0.926	0.236	0.760
CROPCODE Four Streams	0.965	0.968	0.888	0.779	0.288	0.635	0.919	0.304	0.743

Table 9. **Comparison against published pre-trained encoders (E8).** Published self-supervised visual encoders are evaluated on identical held-out test plants using mean-pooled features and the same ridge regression probe protocol.

Dataset / Capacity Variant	Epochs	Sorghum				Maize			
		Chamfer	Height	Roll	Leaf Len	Chamfer	Height	Branch	Stem Rad
Training Dataset Scale (E3, Base Model)									
1,000 Plants	6,169	3.13	0.952	0.716	0.249	3.54	0.393	0.851	0.593
3,000 Plants	2,100	2.86	0.932	0.679	0.213	1.46	0.524	0.867	0.624
10,000 Plants	631	2.48	0.962	0.749	0.245	0.87	0.427	0.853	0.717
10,500 Plants	600	2.53	0.962	0.760	0.249	0.99	0.635	0.919	0.743
Encoder Capacity Scale (E4, Full Dataset)									
ViT-Small (25 M)	600	2.86	0.964	**0.817**	**0.277**	1.22	0.563	0.886	0.682
ViT-Base (114 M)	600	**2.53**	0.962	0.760	0.249	**0.99**	**0.635**	**0.919**	**0.743**
ViT-Large (332 M)	600	2.68	0.948	0.711	0.246	1.06	0.580	0.887	0.614

Table 10. **Dataset scaling and encoder capacity ablations at equal step budgets.** Evaluated at a fixed budget of 197,400 optimizer steps. Chamfer distance ($\times 10^{-3}$) and linear probe R^2 (validation) show that, at fixed step compute, larger encoders do not help and more data helps reconstruction more reliably than it helps the probes.

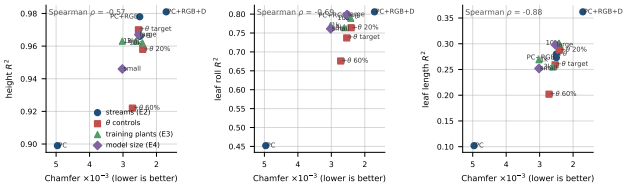


Figure 12. **Divergence between Chamfer distance and frozen-latent probe metrics.** Validation Chamfer distance plotted against height probe R^2 across twelve model variants.

projections of frozen encoder [CLS] tokens. Embeddings form continuous, smooth manifolds structured by global plant scale and organ posture rather than discrete clusters. Latent effective participation ratio is low (4.4–6.3 for sorghum, 6.7–11.3 for maize out of 768 dimensions), indicating a low-dimensional representation.

5. Discussion and Limitations

We discuss the core insights, metric divergences, architectural principles, and limitations established by our experiments.

Privileged parameter visibility and modality competition. A central design question in multimodal architectures trained alongside generative models is where privileged structural information should enter the network. The procedural parameter stream θ is the only modality that states target phenotypic traits verbatim, yet it is never available at deployment in the field. Across our controlled ablations (Sec. 4.6), exposing parameter tokens to the encoder during pretraining degraded the quality of learned sensor representations in both sorghum and maize. In sorghum, every model arm that saw parameter tokens in the encoder learned inferior visual features, with the degradation on the size traits growing with the stream's visibility (Tab. 8).

This behavior can be understood through the lenses of shortcut learning, modality competition [34], and negative gradient transfer [48]. When ground-truth parameter tokens are visible to the transformer encoder, they directly encode target geometry. A plausible account is that the encoder learns a low-cost shortcut through these parameter tokens, starving the visual (RGB, depth) and point cloud streams of gradient updates; when parameter tokens are then absent at inference, the sensor representations prove impoverished.

The species-specific control arms (Tab. 8) reveal an additional nuance:

- **Sorghum**: Keeping parameter tokens strictly as a decoder target ($\rho_\theta = 1.0$) continues to lower sensor latent quality (R^2 drops from 0.977 to 0.966 on height). Sorghum morphological parameters are tightly coupled (effective rank two); a plausible mechanism is gradient competition during multi-task decoding that pulls sensor representations

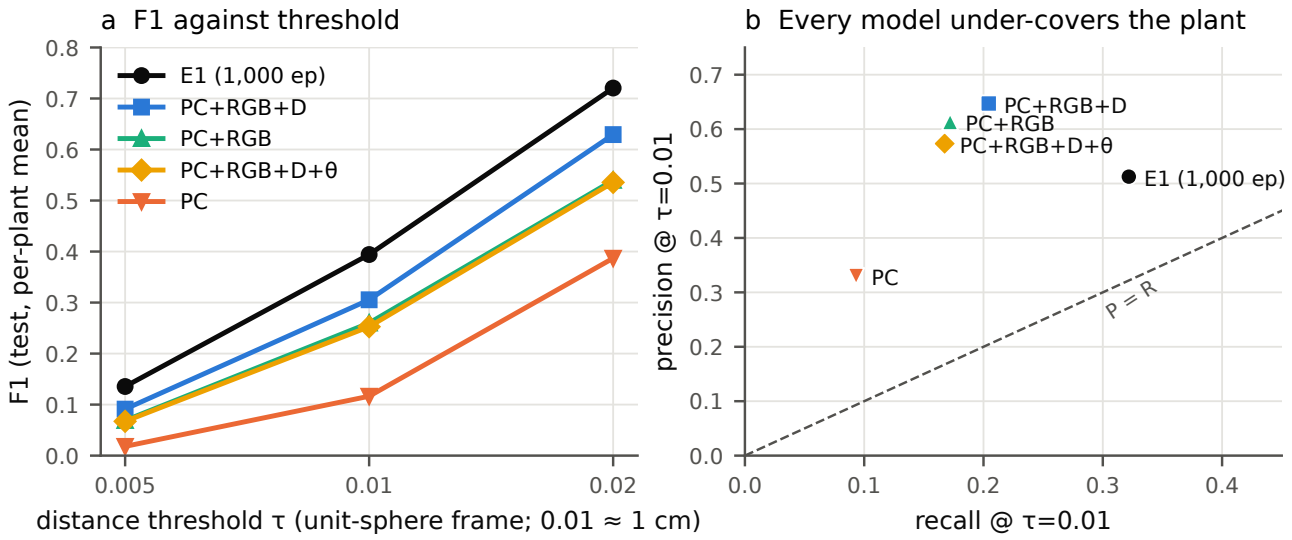


Figure 13. **Point cloud completion precision, recall, and F_1 evaluation (sorghum test).** (a) F_1 score against distance threshold τ. (b) Precision vs. recall at $\tau = 0.01$.

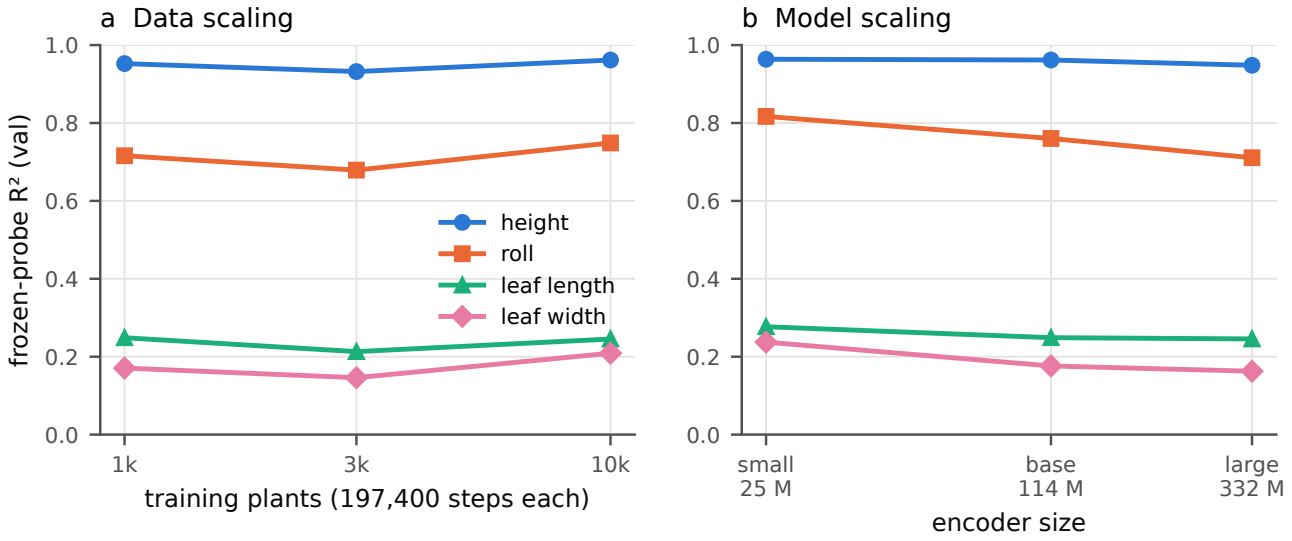


Figure 14. **Dataset and model capacity scaling at equal step compute (sorghum).** Linear probe validation R^2 plotted against dataset size (a) and encoder parameter capacity (b).

toward global plant size, the leading principal component of the sorghum latent.

- **Maize**: Keeping parameters strictly as a decoder target ($\rho_\theta = 1.0$) or restricting visibility to 20% improves sensor latent probing across all eleven traits ($R^2 = 0.776$ vs. 0.633 on height). Maize parameters contain 11 independent morphological dimensions; supervising them as a generation target provides structured multi-task decoder gradients, consistent with a regularizing effect on sensor latents without encoder shortcuts.

These findings yield an architectural design rule for multimodal masked autoencoders trained on synthetic or procedural data: *privileged generative parameters should never be largely visible to the encoder during pretraining*. Setting parameter tokens strictly as a decoder reconstruction target ($\rho_\theta = 1.0$) or applying heavy gating ($\rho_\theta \leq 0.20$) prevents encoder shortcut learning while preserving generative parameter output capabilities; whether the parameters help

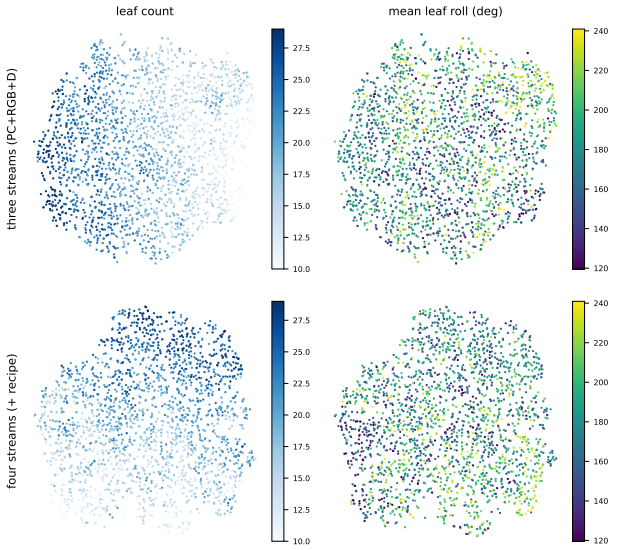


Figure 15. **Latent space visualization (sorghum validation set).** t-SNE projections of frozen encoder `[CLS]` tokens for 3-stream (PC + RGB + D) and 4-stream (+θ) models, colored by plant leaf count and mean leaf roll.

even as a pure reconstruction target depends on the generator.

Metric divergence on slender geometry. A critical empirical finding of this work is the structural divergence between nearest-neighbor point cloud metrics (such as Chamfer distance) and downstream phenotypic trait probing on slender geometry (Sec. 4.7, Fig. 12). Standard Chamfer distance evaluates surface proximity by computing average nearest-neighbor point displacements:

$$\mathrm{CD}(P,\hat{P}) = \frac{1}{|P|}\sum_{p\in P}\min_{q\in\hat{P}}\|p-q\|_2^2 + \frac{1}{|\hat{P}|}\sum_{q\in\hat{P}}\min_{p\in P}\|p-q\|_2^2. \tag{9}$$

Because crop organs (such as maize and sorghum leaves) are thin, high-aspect-ratio surfaces, a predicted point cloud can achieve low Chamfer distance simply by clustering points onto bulky central stalks while omitting entire leaf blades. In contrast, downstream linear probes fitted on frozen encoder embeddings directly measure whether latent representations retain the structural parameter information required for phenotyping.

Throughout our experiments, Chamfer distance and linear probes rank model variants differently:

1. **Depth stream addition**: Adding metric depth to RGB photographs reduces Chamfer distance by 28% in sorghum and 25% in maize, but yields negligible change in downstream probe R^2 scores (Tab. 7). Depth ordering provides spatial surface proximity for decoders without expanding high-level trait information in the encoder.
2. **Data scaling**: Increasing pretraining dataset volume from 1,000 to 10,000 plants reduces sorghum Chamfer error by 21% ($3.13 \rightarrow 2.48 \times 10^{-3}$), whereas sorghum probe scores saturate by 10,000 plants (Tab. 10).
3. **Partial camera visibility**: Restricting point cloud inputs to a single camera perspective increases Chamfer error $2.6\times$ to $6.5\times$, whereas refitted linear probes of the fused arms lose at most 0.030 in height R^2 (Tab. 4).

These disagreements demonstrate that Chamfer distance alone is an insufficient metric for evaluating slender 3D shape reconstruction. We advocate evaluating structural downstream trait probes alongside Chamfer distance, and incorporating distribution-matching losses (such as Sinkhorn optimal transport divergence, Eq. (6)) during training to force decoders to cover full organ geometry rather than collapsing points onto central stalks.

Partial coverage, camera viewpoint, and row occlusion. Sensor observations in agricultural field campaigns are inherently partial. Our experiments analyze three aspects of partial sensing:

- **Viewpoint elevation dominance**: Across isolated camera perspectives, camera elevation angle dominates single-image reconstruction difficulty, accounting for $\eta^2 = 0.24$–0.31 of per-plant reconstruction variance (Fig. 4). Steep top-down and bottom-up perspectives suffer from severe self-occlusion where leaves stack vertically along the optical axis, whereas side-on views expose full blade silhouettes.
- **Covariate shift under partial visibility**: While phenotypic trait information persists in encoder representations under partial surface coverage (refitted 1-camera height $R^2 = 0.960$), directly transferring linear probes trained on 100% complete clouds to partial inputs results in performance failure ($R^2 \rightarrow -0.26$, Fig. 6). Partial visibility induces a systematic covariate shift in latent feature space. Consequently, downstream phenotyping readouts deployed in the field must be refit or calibrated on sensor-matched partial coverage inputs.
- **Implicit row occlusion reasoning**: In planted rows, neighboring foliage accounts for up to two-thirds of input point clouds and pixel areas. Standard models trained on isolated plants suffer catastrophic failure under row occlusion because decoders attempt to incorporate neighboring leaves into target plant reconstructions (Tab. 5). Fine-tuning decoders to predict target main plant geometry behind occluding neighbors removes 92% of neighbor-induced occlusion error in maize and 85% in sorghum, forcing the network to perform implicit occlusion reasoning and segment target organs from inter-plant contact foliage.

Importantly, procedural parameter outputs ($\hat{\theta}$) demon-

strate superior robustness to partial visibility compared to raw point cloud completions. Because the procedural generator $G(\hat{\theta})$ acts as an explicit biological prior, re-rendered procedural meshes generate complete 3D geometry even when input sensor observations capture only a fraction of the plant surface.

Capabilities and boundaries of cross-modal distillation. Cross-modal teacher–student distillation (Sec. 3.4) converts a multimodal pretraining artifact into a fast, deployable single-sensor inference model. Distilling a full-modality teacher (RGB + D + PC + θ) into single-sensor student networks reduces overall cross-modal generation loss by 30.2% in sorghum and 43.5% in maize (Tab. 6). Distillation gains are concentrated on the weakest input modalities (photograph-to-cloud Chamfer error falls by 61%) and the most challenging plant geometries (Fig. 4B). Furthermore, supervising student models across all four decoder targets simultaneously outperforms a specialized parameter-only regression student (first release; no final-data specialist was trained), consistent with auxiliary visual reconstruction tasks acting as regularizing gradients for parameter regression.

However, distillation exhibits clear boundary limits:

1. **Viewpoint robustness**: Distillation lowers absolute reconstruction error across all camera views but does not flatten the U-shaped elevation difficulty curve (Fig. 4A). Viewpoint robustness cannot be acquired purely through distillation; it requires balanced elevation sampling during pretraining data capture (Fig. 5).
2. **Unobservable parameter instability**: Parameterizations submitted to inverse regression must be restricted to traits that field sensors can physically resolve. In early experimental iterations containing unobservable leaf waviness parameters, distillation caused these unconstrained parameter heads to develop severe view-dependent instability without improving trait skill (skill ≈ 0). Restricting procedural parameter representations to observable physiological attributes (insertion heights, blade lengths, widths, angles, and rolls) ensures stable cross-modal distillation.

Limitations and future directions. We acknowledge several limitations in our current study:

- **Single-seed ablation budgets**: Controlled ablation experiments (Tabs. 7–10) were executed under a standardized compute budget of 197,400 optimizer steps using a single seed per arm. Run-to-run baseline variance estimated from near-duplicate runs is $\approx 2\%$ in Chamfer distance and ≤ 0.011 in probe R^2; the non-monotone maize scaling differences, though significant under the paired bootstrap, could still reflect run-to-run variance.
- **Procedural generator expressivity boundaries**: An inherent limitation of inverse procedural modeling is that recovered representations are bounded by the expressivity of the underlying generator G. If a physical plant exhibits morphological deformations unrepresented in G (such as severe disease lesions, structural breakage, or tassel droop, which the generator renders without learnable variation), the framework fits the nearest valid procedural parameter configuration. Expanding procedural generators to support broader developmental malformations represents an important avenue for future research.
- **Synthetic renderer regularities**: Synthetic rendering pipelines can introduce unintended canonical cues (such as fixed leaf plane azimuth alignments) that pose-aware baselines may exploit as shortcuts. Rigorous evaluation on real-world field captures (Sec. 4.8) remains essential to verify true sim-to-real domain transfer.

By releasing FIELDPROC (comprising 30,000 synthetic plants across maize and sorghum with paired multimodal captures, physically simulated row occlusion, and exact procedural parameter ground truth), we aim to provide the 3D vision and digital agriculture communities with a benchmark for evaluating multimodal completion and inverse procedural modeling on slender, heavily occluded geometry.

6. Conclusion

CROPCODE recovers a complete, editable procedural plant from the data of whichever sensor a field campaign used, in one forward pass, by using the generator for both the training data and the output representation. Two findings from its ablations apply beyond plants: the generator's parameters harm the representation when most of them are visible to the encoder, and on slender geometry a downstream probe, not Chamfer distance, should decide what a modality is worth. A single camera's partial view requires refitting the readout and leaves a measured transfer gap; the row-occluded and real-scan evaluations will show whether the generator's prior covers the remaining difference to field data. Sharpen once E10 lands.

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Procedural Reconstruction of Slender Plants from Any Field Modality

Supplementary Material

A. Detailed Literature Review

This appendix expands the compressed related-work discussion of Section 2 to the six bodies of work on which the paper builds or from which it departs.

A.1. Procedural plant models and inverse procedural modeling

Procedural modeling of plants began with L-systems [35] and matured into generators whose parameters are organ-level, biologically meaningful quantities [19, 20]. Inverting such models, that is, recovering their parameters from observations, has a long history for woody species: Stava et al. fit tree parameters by stochastic search [43], Talton et al. formulated procedural inversion as MCMC inference [44], and Guo et al. inferred L-systems from point clouds [17], with neural variants following [28]. These methods optimize or infer per instance, and none performs amortized, feed-forward inversion of a plant generator from field-degraded input. The closest amortized precedents are outside agriculture: shape-program induction [45], diffusion-based inverse procedural content generation from a single image [53], and cadrille, which maps images, point clouds, or text to executable CAD programs [24, 26]. All operate on rigid, isolated, clean-input objects; none addresses deformable slender organs, inter-plant occlusion, or real-sensor noise, and none studies whether the program parameters should participate in the encoder.

A.2. Learned parametric recovery of crop plants

Two 2025 systems recover parametric plants from scans. Demeter learns a morphology model (topology, articulation, PCA shape) from real soybean point clouds [8]; Ghrer et al. infer grammar-style parameterized plants from 3D scans [14]. Both take a single, reasonably complete point cloud as input and neither addresses missing modalities or row occlusion. CropCraft recovers crop structure by Bayesian optimization of a procedural model against multi-view NeRF depth [52]; the approach is accurate but per-scene and slow. A staged pipeline annotates terrestrial laser scans with a vision-language model and fits a NURBS procedural descriptor by differentiable refinement [9, 18, 19]; it handles isolated, occlusion-free plants at $\sim$10 minutes each and serves in this paper as both the occlusion-free baseline and the source of real-data procedural ground truth. Tree-D Fusion reconstructs simulation-ready urban trees from street imagery via diffusion priors [29], but its envelope-then-branch pipeline targets woody silhouettes, a different geometry class from thin cereal blades.

A.3. Point-cloud completion and slender structures

Learned completion runs from PCN [51] through transformer approaches [49, 50] to recent extrapolation–interpolation designs [5], trained and evaluated on benchmarks of compact rigid objects. Neither PCN's eight categories nor ShapeNet-55/34 contains a slender-under-occlusion class, and completion systems assume the missing geometry is self-occluded surface, without clutter from *other* objects inside the crop region. Plant-specific completion is emerging, with amodal segmentation plus completion for occluded plants [27], but it outputs geometry rather than structure or parameters. Work on curve-aware and thin-structure reconstruction [13, 38] addresses the slender regime, in which the standard metric itself fails: Chamfer distance under-weights thin sheets [47], a failure now documented analytically [42] and addressed by density-aware and information-theoretic variants [30, 47]. We further compare geometric metrics with frozen-latent phenotype probes and show that they can rank modalities differently.

A.4. 3D plant phenotyping

Organ-level 3D phenotyping datasets, including Pheno4D's labeled maize and tomato sequences [41], Crops3D [54], and field-scale maize scans with procedural fits [25], support trait extraction from point clouds [36, 37]. Radiance-field capture is increasingly practical in field conditions [2]. No existing dataset, however, combines real field capture, slender organs, inter-plant row occlusion, and procedural ground truth, and our benchmark is designed to combine all four.

A.5. Multimodal masked pretraining and distillation

Masked autoencoding [21] extends to multiple modalities: MultiMAE over RGB, depth, and semantics [3]; 4M and 4M-21 to any-to-any generation over tokenized modalities [4, 31]; Point-MAE to geometry [33]; and Embodied-MAE, on which our architecture is based, to RGB, depth, and point clouds for robot manipulation [10]. Missing-modality robustness is an active concern [40], and cross-modal distillation from multi-sensor teachers to single-sensor students is established for discriminative tasks; we distill a *generative* full-modality teacher for structured-parameter regression, and encode continuous parameters with xVal-style number tokens [15]. Our finding that a privileged parameter stream suppresses representation quality connects to modality competition [34] and gradient conflict [7, 48], which offer candidate mechanisms; no prior

work, to our knowledge, documents the effect for generator parameters in multimodal pretraining.

A.6. Synthetic-data pretraining and sim-to-real

Pretraining without natural data is viable, from abstract fractals [23] to procedural worlds [16, 39], and domain randomization reduces the reality gap [46]. The value of synthetic data is known to be task-dependent [11], and self-consumption is a documented risk [1]. Our setting is the favorable case: the deployment target *is* the generator's parameterization, the generator spans the morphology of the crop, and the method is evaluated on real field data.

B. Detailed Ablation Studies

All numbers in this appendix are on the final data releases of both species unless marked as first release. Protocol: every arm trains for 197,400 optimizer steps at global batch 32; representation quality is a ridge probe on the frozen CLS token, one fixed view per plant, identical target vectors across arms, with paired bootstrap intervals over plants (a win or a loss needs the 95 % interval to exclude zero on both val and test); generation quality is the validation Chamfer at the matched checkpoint.

B.1. Modality value-add (E2)

Eight pretraining arms per species differ only in which token streams exist (every stream set that contains the cloud).

B.2. Token-budget control arms

In the four-stream arm the three sensor streams share the visible budget with the parameter tokens (107.4 visible sensor tokens against 117.0 in the three-stream arm, parameter tokens 58 % visible). The control arms keep the parameter tokens outside the shared budget, restoring the 117.0 sensor tokens, at three parameter visibilities (20 %, 60 %, 0 % = target only). In sorghum no visibility recovers the three-stream arm: height 0.959–0.966 against 0.977, roll 0.702–0.771 against 0.781, and the 60 % and target-only arms lose all nine learnable traits on both splits. In maize the 20 % and target-only arms beat both the shared-budget arm and PC+RGB+D on all eleven traits, while the 60 % arm loses nine. Full table and intervals: Tab. 8 of the main paper.

B.3. Parameters as a target only

Reconstructing the parameters without ever encoding them (100 % masked as input) still loses to omitting the stream in sorghum: size -0.011, roll -0.07, mean leaf length -0.10, leaf width -0.15 (val), intervals clear of zero. Its latent is the most size-dominated of the sorghum arms: the first principal component carries 44 % of the variance and correlates with leaf count at $|r|{=}0.95$, and the participation ratio falls to 4.3 against 6.7 for PC+RGB+D. In maize the same arm is the best four-stream variant and beats PC+RGB+D on every trait, so the effect of the reconstruction target depends on the generator.

B.4. Data and model scaling at fixed compute

B.5. Distillation, per source modality

From the headline model of each species, distillation lowers the mean generation loss by 30.2 % in sorghum and 43.5 % in maize (val, epoch-0 warm start against epoch 100). The gain is largest where the warm start is weakest: sorghum Chamfer from the point cloud falls from 5.48 to 1.39×10^{-3} and from the parameters alone from 22.7 to 6.83; maize Chamfer from the photograph falls from 11.3 to 0.77. The parameter error from a point cloud alone halves in sorghum ($0.040 \rightarrow 0.020$) and changes little in maize ($0.020 \rightarrow 0.018$). On the first release, distilling all four targets beat distilling the parameter head alone ($-43.8\,\%$ against $-36.9\,\%$ parameter error), and distillation moved the whole view-difficulty curve down without flattening it (9 of 10 views improved). LR caveat (first release): the parameter specialist ran at 3×10^{-5} against 10^{-4}.

B.6. View invariance without labels (first release)

Ten views of one plant share one parameter ground truth, so the spread of ten predictions is a label-free view-invariance measure. Distillation roughly halves error *and* view spread on the fields with skill (starting point $-59\,\%$, branching angle $-60\,\%$). Zero-skill fields instead become *more* view-dependent, as the head starts to track image noise; the final generator drops those fields. View stability and accuracy are separate properties: several fields move less than 0.3 of their spread across views while remaining worse than predicting the plant mean.

B.7. Frozen-encoder baselines and camera-pose forensics

On sorghum the three-stream arm beats every published frozen encoder (EmbodiedMAE-B [10], Point-MAE [33], DINOv2 [32], MultiMAE [3]) on every trait, and EmbodiedMAE-B, the strongest, on eight of nine paired over plants (size $+0.004$ to $+0.007$, roll $+0.04$, leaf length $+0.08$, leaf width $+0.09$); it also beats a network of our architecture trained from scratch on the probe targets on seven of nine traits. On maize the four-stream arm beats EmbodiedMAE-B on 10 of 11 traits and the best pose-free baseline per trait on 9 (val, point estimates), but the maize network trained from scratch is ahead of every frozen 600-epoch arm on all eleven; the maize headline model closes most of that gap. On the first maize release, supplying Point-MAE with each view's full camera pose gave it the generator's fixed plant azimuth, a canonicalization that no field rig provides; against Point-MAE given gravity alone (the information a calibrated rig provides), our pose-free

Arm	Sorghum Chamfer$\times 10^{-3}$	Sorghum height R^2	Maize Chamfer$\times 10^{-3}$	Maize height R^2
PC	5.00	0.895	2.63	0.171
PC + depth	2.45	0.976	2.55	0.453
PC + RGB	2.57	0.979	1.61	0.620
PC + RGB + depth	1.85	0.977	1.21	0.633
PC + params	4.84	0.860	2.49	0.256
PC + depth + params	4.08	0.907	2.54	0.382
PC + RGB + params	3.00	0.962	1.21	0.509
all four	2.53	0.962	0.99	0.635

Table 11. E2, val. In sorghum the parameter stream raises Chamfer in every set except PC alone and lowers the probe; in maize it lowers Chamfer but its effect on the probe depends on the set (added to PC: 7 wins, 0 losses of 11; to PC+RGB: 3 wins, 5 losses; to PC+RGB+D: a draw). Full probe columns are in the main E2 table.

	Sorghum Chamfer$\times 10^{-3}$	Sorghum height R^2	Maize Chamfer$\times 10^{-3}$	Maize height R^2
1,000 plants	3.13	0.952	3.54	0.393
3,000 plants	2.86	0.932	1.46	0.524
10,000 plants	2.48	0.962	0.87	0.427
10,500 (all)	2.53	0.962	0.99	0.635
small (25 M)	2.86	0.964	1.22	0.563
base (114 M)	2.53	0.962	0.99	0.635
large (332 M)	2.68	0.948	1.06	0.580

Table 12. Four-stream arm, 197,400 steps per row, val. Paired over plants in sorghum, the 10,000-plant arm ties the full-data arm (one win, no loss), the 3,000-plant arm loses six traits and the 1,000-plant arm four. In maize every reduced-data arm loses 10 of 11 traits to full data and the probe is not monotone in data (the 3,000-plant arm beats the 10,000-plant arm on height, leaf length and leaf count). Base is the best size in both species; in maize small and large each lose 10 of 11 traits to base.

model won 5 of 9 decided targets. Tassel droop is the weakest trait for every encoder of this design.

B.8. Latent structure (E9)

The latents are continua, as expected for a single-generator population: best silhouette 0.15 to 0.24 at k=2, adjusted Rand index $\approx$ 0 against shape deciles, t-SNE trustworthiness 0.95 to 0.99. The participation ratio is 4.4 to 6.3 (of 768) in sorghum and 6.7 to 11.3 in maize, consistent with maize's more independent generative factors, and the dominant principal component of the sorghum latents is the plant's size. Linear decodability is distinct from local embedding structure: the probe reads roll at R^2 0.78 (PC+RGB+D, val) from a latent in which roll is barely organized locally. We therefore use probes, not t-SNE smoothness, for representation claims on continuous generators.

C. Dataset Generation

FIELDPROC holds four synthetic sets: isolated single-plant captures of maize and sorghum, and *row views* in which the same plants stand in a crop row between two neighbors and are photographed by a ground rover. All images are 1024×1024 with a 40° vertical field of view ($f_x = f_y = 512/\tan 20^\circ = 1406.7$ px, principal point at the center), in the OpenGL convention (y up, the camera looking down $-z$).

C.1. Procedural plants

Maize. Plant i of 15,000 is generated with seed i by a procedural maize generator whose traits are drawn from Gaussians (height 0.832 ± 0.104 m, leaf count 12.35 ± 2.57, base leaf length 0.448 ± 0.057 m, base leaf width 0.066 ± 0.010 m, base leaf angle $43.98^\circ \pm 10.13^\circ$), coupled through four latent traits (vigor, slenderness, posture, texture) with correlation strength 0.735, with bottom, middle and top canopy curves and per-leaf jitter (scale 0.687). The plants are 0.62–1.44 m tall (median 1.01 m). The generator record is reduced to the plant height, the leaf count and, per leaf, its distance along the stem, length, width, azimuth, angle and droopiness; per-leaf curvature is added by rebuilding the midrib B-spline (target elevation angle$+$droopiness$\cdot t^2$) and integrating $\theta = \int |\kappa|\, ds$ over the blade, $s \in [0.22, 1]$, with the blade length S from 10-node Gauss–Legendre quadrature.

Sorghum. Plant i of 15,000 is generated with seed i by the EvoEngine sorghum generator. It draws the plant heading uniformly in $[0°, 360°)$, the stem tilt ($0 \pm 1.5°$), the leaf count (20 ± 2, clamped to $[2, 128]$) and the internode length (0.04 ± 0.03 m), and sets the stem length to internode $\times\, n_{\text{leaves}}/(1 - s_0)$, with s_0 the first leaf's starting point. Leaf k at $t = k/(n-1)$ takes its starting point, length, width, roll ($\text{roll}_k = (k \bmod 2) \cdot 180° + \text{roll}(t)$), branching angle, bending, curling and waviness from curves over t. No panicle is generated. The plants have 4–35 leaves (mean 19.5) on stems of 0.19–1.74 m.

C.2. Splits

Both species are split by plant into 10,500 / 2,250 / 2,250 plants. *Maize:* 19 scalar features are robustly standardized, $z = (x - \text{median})/(1.4826\,\text{MAD})$, each plant is scored by $\sqrt{z^\top(\Sigma + 10^{-6}I)^+ z}$, and plants are dealt from most to least extreme alternately to test and validation until both hold 2,250 (seed 0). Train scores span 1.56–4.60, validation 4.60–23.3, test 4.60–32.2. *Sorghum:* the extremeness reproduces, to the stored precision, as the Mahalanobis distance from the mean (sample covariance) over leaf count, stem length, mean and maximum leaf length, and mean and standard deviation of leaf roll. The hold-out probability rises with a plant's extremeness rank, from 0.3 % in the least extreme 5 % to 70 % in the most extreme 5 % (median rank 0.75 on validation and test, 0.37 on train). This rule is inferred from the released assignment; the script that wrote it was not found.

C.3. Isolated-plant captures

Both species are rendered off-screen with `pyrender`. The camera looks at the plant's bounding-box center $\mathbf{c}$ with world up $(0, 1, 0)$; lighting is ambient 0.15 plus three white directional lights aimed at $\mathbf{c}$ from $(1, 1, 1)$, $(-1, 0.5, -0.5)$ and $(0, 1, 0)$ (intensities 4.0, 2.0, 1.5), on a gray (0.5) background. Depth z with near and far planes n, f is stored as $v = \lfloor \text{clip}((z - n)/(f - n), 0, 1)(2^{32} - 1) \rfloor$ across the four 8-bit channels of a PNG, most significant byte first; $v = 0$ is background. A pixel (u, v) with depth $z > 0$ back-projects to

$$\mathbf{p}_{\text{cam}} = \left(\tfrac{u - c_x}{f_x} z,\ -\tfrac{v - c_y}{f_y} z,\ -z \right), \tag{10}$$

and clouds are drawn uniformly without replacement from these points.

Maize: ten view directions per plant are drawn uniformly on the sphere (seeded per plant), at distance $1.05\, r / \sin 20°$ with r half the bounding-box diagonal; a fourth light (1.5) is placed at the camera, and textured leaf, stem and tassel materials are double-sided. Each view stores RGB, an RGBA image with the background cleared, depth, the camera pose, the trimmed parameter file, and an 8,192-point cloud back-projected from that view's depth (Eq. 10), so the cloud contains only the surface visible to the camera. *Sorghum:* the ten directions form a Fibonacci sphere, $y_i = 1 - 2(i + \frac{1}{2})/10$, $\phi_i = i(\sqrt{5} - 1)\pi + \epsilon_i$ with a small fixed jitter, identical for every plant (five above and five below the horizon), at 2.5 m; one leaf texture covers the plant and there is no camera light. Each view stores RGB, depth, the camera pose, the parameter file, and the complete, occlusion-free labeled plant cloud moved into the camera frame.

C.4. Row views

Composition. Within a split with plant list L, row i has main plant L_i, right neighbor L_{i+1} and left neighbor L_{i+2} (indices modulo $|L|$), so every plant is a main plant once. The world frame is the main plant's (stem base at the origin, row along x, rover on the $+z$ side), with the neighbors' stem bases at $(\pm d, 0, 0)$, $d = 0.15$ m. Each row is rendered at ten orientations: the main plant is turned about its stem base by $\theta \in \{0°, 10°, \ldots, 90°\}$, each neighbor by $\theta + \delta$, $\delta \sim U(-15°, 15°)$ (seeded per split and row). Sorghum stems lean, so each sorghum neighbor is also rotated about its stem base by the smallest rotation that makes its stem axis (the principal axis of its stem vertices) parallel to the main plant's; the main plant is never tilted, and maize is not tilted.

Contact solver. The three plants are made free of intersections while staying as close as possible to their grown shapes; they may touch but not cross. Each blade is a chain of 8 cm segments from collar to tip whose joints bend along circular arcs, up/down and sideways, so the blade stays smooth, keeps its length and stays attached. Stems and sheaths (everything within 2.5 cm of the stem axis) and the maize tassel are rigid. A leaf's joint bends sum to a total-turn limit, and a leaf may not bend into its own stem or cross its own plant's leaves more than it does as grown. Where a leaf would cross another plant, it bends at the joint before the contact by the smallest clearing angle, smallest first, so it comes to rest on or under what it meets. Neighbors bend first; a main-plant leaf bends only where a neighbor cannot move (a neighbor's stem or sheath, or a neighbor leaf with no clear shape). A sorghum neighbor leaf that cannot bend clear may also roll about its stem by up to $\pm 90°$. Candidates are screened on 1 cm occupancy grids and every remaining contact is decided by exact triangle–triangle tests; a found layout is then settled, each bent leaf easing back toward its grown shape to the least bend that still clears. Unsolved layouts are retried under looser limits (Tab. 13), then at spacings raised in 5 cm steps up to 1.5 m and refined in 1 cm steps.

Independent intersection check. Solved layouts are verified by a check that shares no code with the solver: a sepa-

rate NumPy implementation takes the plant meshes at their rendered positions and, for every pair of plants in the row, runs exact triangle–triangle intersection tests between their triangles. A row passes only if no triangle of one plant intersects a triangle of another. The check found no crossing plant pairs on the rows it was run on; the coverage of this check is not recorded; re-run it over all 300,000 shots and state the count here.

Stage	collar side	blade side	up/down	total turn
0	30°	10°	45°	90°
1	60°	10°	60°	120°
2	90°	30°	70°	180°

Table 13. Staged bend limits of the contact solver (per joint, except the total turn per leaf).

Rover camera and outputs. One shot is taken per orientation from $(0, h, \mathrm{dist})$, looking horizontally at $(0, h, 0)$ with h the height of the main plant's bounding-box center; $\mathrm{dist} = 1.0$ m for maize and 2.5 m for sorghum, near 0.05 m, far $\mathrm{dist} + 2$ m, with the lens, lights and materials of the isolated captures. Each shot is rendered twice from the same camera: the input with all three plants (RGBA, depth, and a 24,576-point cloud of every visible plant) and the ground truth with the main plant alone in its bent shape (RGBA, depth, 8,192 points), with the main plant's parameter record. The maize input clouds were first produced at 8,192 points and rebuilt at 24,576 from the stored depth; the old clouds lie within 8.4×10^{-8} m of the rebuilt ones.

Split	shots	at d=0.15 m	stage 0 / 1 / 2	max d
train	105,000	76.4 %	81,679 / 16,138 / 7,183	0.33 m
val	22,500	78.3 %	17,780 / 3,321 / 1,399	0.28 m
test	22,500	79.0 %	17,824 / 3,284 / 1,392	0.30 m

Table 14. Maize row statistics. The 99th percentile of the spacing is 0.23 m in every split, and no orientation failed. On 10 sample sorghum rows (100 orientations), 85 were solved at 0.15 m and the rest at 0.16–0.21 m.

Caveats. The row ground truth is the main plant as bent in that row, while its parameter record describes the grown plant; main-plant leaves are bent only where unavoidable. At 2.5 m the lens covers about 1.8 m of height, so the top or base of tall sorghum plants can leave the frame (in the 10 sample sorghum rows the main plant touches the top edge in 58 of 100 shots). The two isolated sets differ in view sampling, camera distance, materials, lighting and per-view cloud. Sorghum plant 0 uses seed 0, for which the generator does not reseed. Check that plant 0's parameters match its mesh.